

Protecting a bosonic qubit with autonomous quantum error correction

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Jeffrey M. Gertler¹, Brian Baker², Juliang Li¹, Shruti Shirol¹, Jens Koch² & Chen Wang^{1✉}

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To build a universal quantum computer from fragile physical qubits, effective implementation of quantum error correction (QEC)¹ is an essential requirement and a central challenge. Existing demonstrations of QEC are based on an active schedule of error-syndrome measurements and adaptive recovery operations^{2,3,4,5,6,7} that are hardware intensive and prone to introducing and propagating errors. In principle, QEC can be realized autonomously and continuously by tailoring dissipation within the quantum system^{1,8,9,10,11,12,13,14}, but so far it has remained challenging to achieve the specific form of dissipation required to counter the most prominent errors in a physical platform. Here we encode a logical qubit in Schrödinger cat-like multiphoton states¹⁵ of a superconducting cavity, and demonstrate a corrective dissipation process that stabilizes an error-syndrome operator: the photon number parity. Implemented with continuous-wave control fields only, this passive protocol protects the quantum information by autonomously correcting single-photon-loss errors and boosts the coherence time of the bosonic qubit by over a factor of two. Notably, QEC is realized in a modest hardware setup with neither high-fidelity readout nor fast digital feedback, in contrast to the technological sophistication required for prior QEC demonstrations. Compatible with additional phase-stabilization and fault-tolerant techniques^{16,17,18}, our experiment suggests quantum dissipation engineering as a resource-efficient alternative or supplement to active QEC in future quantum computing architectures.

Robustness in modern classical computers is often aided by passive dissipation acting as a restoring force against environmental perturbations, rather than active error correction protocols. A quantum analogy for dissipative stabilization of qubits, or autonomous quantum error correction (AQEC), is possible but has important distinctions from its classical counterpart (Fig. 1a). Since qubit states form a continuous two-dimensional encoding manifold $\mathcal{C}$, the restoring dissipation must be tailored perpendicular to $\mathcal{C}$ in order not to disrupt the manifold of code states, which leaves any perturbations tangential to $\mathcal{C}$ unrecoverable. Therefore, the encoding manifold $\mathcal{C}$ itself also has to be specifically designed to ensure that the physical errors only occur orthogonal to the surface, as formulated by the Knill–Laflamme QEC criteria¹⁹.

These simultaneous requirements dictate that the form of dissipation needed for AQEC must be exotic²⁰. Advances in quantum reservoir engineering (dissipation engineering)²¹ have paved the way to synthesize dissipation operators not naturally available, enabling stabilization of various non-classical states^{22,23,24}. More recently, dissipation has been tailored to confine a quantum harmonic oscillator to delocalized regions in phase space²⁵ to suppress bit-flips²⁶. However, the stabilized manifold does not permit QEC of the dominant decoherence process in the system: single photon loss. Encouraging proposals for AQEC have emerged for a number of experimental platforms^{10,11,12,13},

but they require hardware architectures yet to be developed. In this Article, we introduce and implement an AQEC scheme in a common transmon-based superconducting circuit QED device^{27,28,29}, which is solely enabled by the synthesis of a highly specific dissipation operator.

Error correction code and strategy

Following the accelerating progress in bosonic QEC research^{30,31}, we take advantage of the large Hilbert space and the long coherence time²⁸ of microwave-photon states in a superconducting cavity to store a logical qubit. This logical qubit, $|\psi\rangle = x|0_L\rangle + y|1_L\rangle$ (with x and y being complex coefficients containing the logical information), is encoded in an odd-parity subspace of the cavity using the following Truncated 4-component Cat (T4C) code³²:

$$|0_L\rangle = C_1|1\rangle + C_5|5\rangle, \quad |1_L\rangle = C_3|3\rangle + C_7|7\rangle, \quad (1)$$

where $C_1, C_3, C_5, C_7 = \sqrt{0.35}, \sqrt{0.9}, \sqrt{0.65}, \sqrt{0.1}$ are code word coefficients of the four Fock-state components, chosen to approximately balance the average photon numbers in $|0_L\rangle$ and $|1_L\rangle$ ($\bar{n} \approx 3.5$). The $\hat{x}$ basis states (instead of $\hat{z}$) of the code, $|\pm_L\rangle = (|0_L\rangle \pm |1_L\rangle)/\sqrt{2}$, resemble the Schrödinger cat superpositions of coherent states, $|\mathcal{C}_\alpha\rangle = (|\alpha\rangle - |-\alpha\rangle)/\mathcal{N}$.

¹Department of Physics, University of Massachusetts Amherst, Amherst, MA, USA. ²Department of Physics and Astronomy, Northwestern University, Evanston, IL, USA. ✉e-mail: wangc@umass.edu

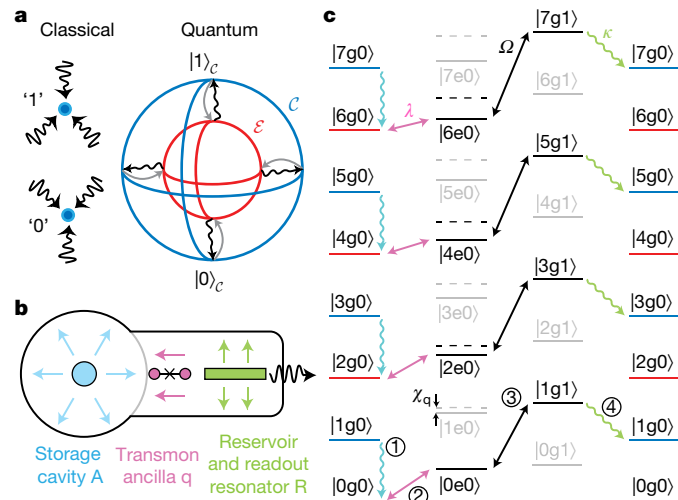


Fig. 1 | Autonomous quantum error correction: concept and protocol. **a**, A classical bit is stored in isolated points in configuration space, which are local energy minima stabilized by dissipation in all directions. In comparison, a logical qubit $|\psi\rangle$ is encoded in a continuous two-dimensional code space $\mathcal{C}$ designed in a way that both natural errors (which map the states to the error space $\mathcal{E}$) and engineered dissipation are only allowed perpendicular to $\mathcal{C}$. **b**, Schematic of the circuit QED device composed of storage cavity A, transmon ancilla q and reservoir resonator R. **c**, AQEC scheme against single-photon loss illustrated in a level diagram (not to scale). The level indices refer to the quantum Fock states of A, q and R, sequentially. A continuous-wave 'transmon comb' is applied to resonantly excite the transmon ancilla with a Rabi rate λ (magenta arrows), selectively targeting the four even-parity states (red levels) when $\chi_q \gg \lambda$. Similarly, a continuous-wave 'mixing comb' targets the $|2n, e, 0\rangle \leftrightarrow |2n+1, g, 1\rangle$ transitions with an equal Rabi rate Ω (black arrows). Both combs are composed of four tones equally spaced by $2\chi_q$, as indicated by the varying slopes of the magenta and black transitions. Spontaneous decay of the reservoir R converts the quantum state back to the code space (blue levels) without leakage of which-path information; note the identical slopes of the light-green arrows. The numbered labels 1–4 show the error and recovery sequence for one of the four parallel paths (using the initial state $|lg0\rangle$ as an example).

The coherent-state amplitude, $|\alpha| \approx 1.87$, is a measure of the size of the cat-state encoding. Single-photon loss, the dominant intrinsic error in a superconducting cavity to be corrected, converts odd-numbered Fock components to even-numbered ones. The photon-number parity $\hat{P} = e^{i\pi\hat{a}^\dagger\hat{a}}$ effectively plays the role of a QEC stabilizer operator, which can be repetitively measured in active QEC protocols^{5,6}.

The main technical accomplishment of this work is the realization of parity recovery by selective photon addition (PreSPA) via a constantly applied dissipation operator

$$\hat{\Pi} = |1\rangle\langle 0| + |3\rangle\langle 2| + |5\rangle\langle 4| + |7\rangle\langle 6|. \quad (2)$$

This operator stabilizes the photon-number parity, by converting even to odd states, while preserving coherence between the Fock components. Whenever a parity jump arises from photon loss, PreSPA performs AQEC by automatically adding a photon back to the cavity. PreSPA is not constructed from a true parity operator ($\propto e^{i\pi\hat{a}^\dagger\hat{a}}$). However, by explicitly constructing a superposition of four targeted dissipative processes, it achieves a workaround to the well recognized challenge of implementing continuous quantum non-demolition parity projection^{13,33}, which has been a major obstacle for AQEC. It should be noted that $\hat{\Pi}$ does not fully reverse the effect of a photon loss event ($\hat{\Pi}\hat{a} \neq \hat{a}$); rather, it leads to a net gain in mean photon numbers for all logical states, which is approximately equivalent to an increase of $|\alpha|$. PreSPA also does not correct for the continuous decrease of $|\alpha|$ in the

absence of parity jumps⁵. However, the accumulated entropy in the size uncertainty of the cat-state encoding is not associated with substantial leakage of logical information (which is encoded in the phases of the cat states), and can be removed from the logical qubit effectively using a proper decoding transformation. Theoretically, instantaneous and exact $\hat{\Pi}$ operations to this T4C code can reduce its logical error rate from single-photon loss by approximately 50 times (or more if a larger $|\alpha|$ is chosen for the encoding). See Methods for the theory of this approximate AQEC protocol.

AQEC technique and device implementation

Our experiment is carried out in a 3D-planar hybrid circuit QED architecture²⁹ at a base temperature of about 10 mK. A high-coherence cylindrical post cavity A (with $T_{1A} = 520 \mu\text{s}$, $T_{2A} = 380 \mu\text{s}$) is used to store the logical qubit²⁸. A dispersively coupled transmon²⁷ qubit q (with $T_{1q} = 39 \mu\text{s}$, $T_{2q} = 17 \mu\text{s}$) is used as an ancilla for encoding and decoding of the cavity state. A coaxial stripline resonator R with fast decay rate ($\kappa/2\pi = 0.58 \text{ MHz}$) is used both for readout and as the source of dissipation (Fig. 1b). The leading-order terms of the system Hamiltonian are:

$$\frac{\hat{H}}{\hbar} = \omega_A \hat{a}^\dagger \hat{a} + \frac{\omega_q}{2} \hat{\sigma}_z + \omega_R \hat{r}^\dagger \hat{r} - \frac{\chi_q}{2} \hat{a}^\dagger \hat{a} \hat{\sigma}_z - \frac{\chi_R}{2} \hat{r}^\dagger \hat{r} \hat{\sigma}_z, \quad (3)$$

where $\hat{a}$ and $\hat{r}$ are the lowering operators in A and R, and $\hat{\sigma}_z$ is the Pauli operator of the transmon. The dispersive shift between modes q and A is $\chi_q/2\pi = 1.32 \text{ MHz}$ and between modes q and R is $\chi_R/2\pi = 2.8 \text{ MHz}$.

The PreSPA operator is implemented with a fourfold degenerate two-stage pumping process, as illustrated in Fig. 1c. Two continuous-wave frequency combs, each consisting of four tones equally spaced in frequency by $2\chi_q$, are applied to drive transitions targeting the four even-number Fock states, $|2n\rangle_A$ ($n=0, 1, 2, 3$). Under the rotating wave approximation, the drive Hamiltonian is

$$\frac{\hat{H}_{\text{drive}}}{\hbar} = \sum_{n=0}^3 (\lambda |2n, e, 0\rangle\langle 2n, g, 0| + \Omega |2n+1, g, 1\rangle\langle 2n, e, 0|) + \text{h.c.} \quad (4)$$

where h.c. is the Hermitian conjugate. The selectivity on individual levels relies on the photon-number-dependent transition frequencies of the dispersive Hamiltonian. Hence, it is critical that $\lambda \ll \chi_q$, so the odd-parity code space remains unperturbed. The other rates follow a hierarchy of $\lambda < \Omega < \kappa < \chi_q$. When a photon emission occurs in R, the even-parity states are simultaneously projected to the odd-parity subspace by gaining a photon. By adiabatic elimination of the fast dynamics in q and R, we obtain $\hat{\Pi}$ as the effective operator acting on cavity A. Phase coherence among the four converted states is expected to persist, since no which-path information leaks into the environment as long as (1) the transition rates along the four paths are identical, and (2) the frequency of the emitted reservoir photon is independent of the path choice. This path-degenerate pumping process is inspired by several theoretical proposals^{11,14,34}, but our two-stage construction of the coherent drives is crucial for achieving path-independent reservoir emission frequencies while maintaining parity selectivity.

Each of the two frequency combs is generated by single-sideband modulation of a microwave carrier source with four intermediate-frequency control signals digitally combined. The amplitudes and phases of these intermediate-frequency signals can all be independently tuned. The ancilla state is measured using dispersive readout via mode R, which is boosted by a travelling wave parametric amplifier³⁵ but not sufficiently optimized to enable single-shot state assignment. Measurement outcomes are converted to a nominal excited state probability by scaling the averaged demodulated signal relative to the reference ground and excited states. After a series of calibration experiments and compensating for multi-tone parametric

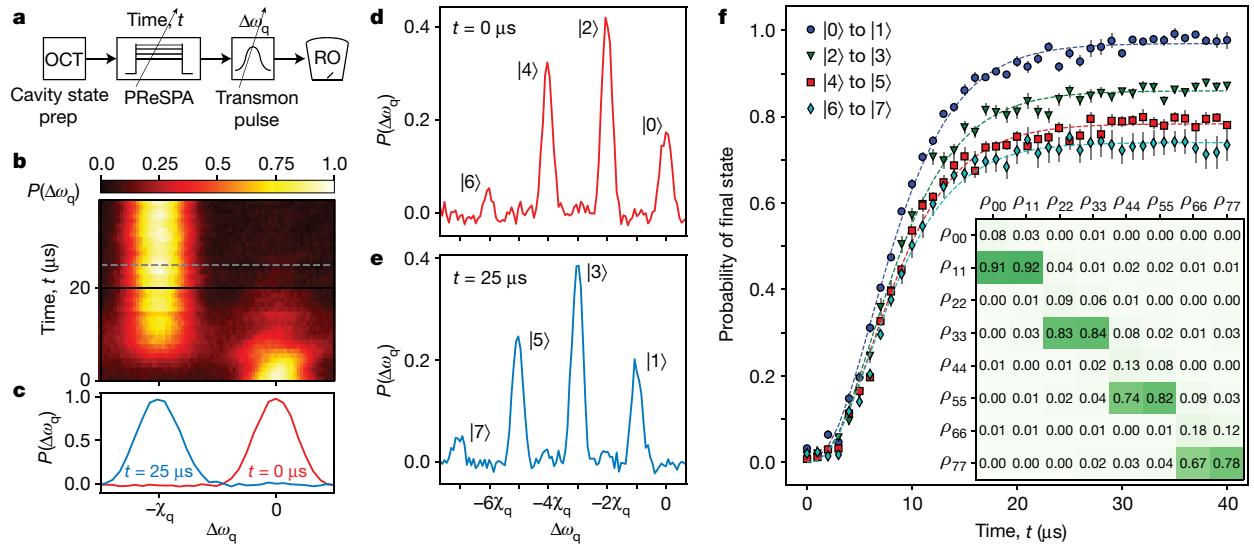


Fig. 2 | Characterization of the PReSPA operator: photon population conversion. **a**, Control pulse sequence of a transmon spectroscopy measurement to infer the cavity photon distribution after PReSPA. We initialize cavity A to a specific initial state using an optimal control theory (OCT) pulse⁴⁶, apply PReSPA for a variable time t , apply a spectrally selective π -pulse to transmon q at a variable detuning $\Delta\omega_q$, and measure the transmon excitation probability $P(\Delta\omega_q, t)$ by performing readout (RO). **b**, Transmon spectroscopy data $P(\Delta\omega_q, t)$ for cavity A initialized in vacuum. The bright feature is shifted from $\Delta\omega_q = 0$ to $-\chi_q$ over time, showing the $|0\rangle_A \rightarrow |1\rangle_A$ conversion. (For $0 < t < 20 \mu\text{s}$ an additional delay time of $20 \mu\text{s}$ is inserted between PReSPA pumps and the transmon π -pulse to improve clarity of the spectroscopy data by allowing the partially excited transmon to relax.) **c**, Cuts of **b** at $t = 0$ and $25 \mu\text{s}$ (grey dashed

line). **d**, **e**, $P(\Delta\omega_q)$ for cavity A initialized in an even-parity cat state at: **d**, $t = 0$ and **e**, $t = 25 \mu\text{s}$. All four spectroscopy peaks corresponding to even photon numbers (red) are shifted by $-\chi_q$ after $t = 25 \mu\text{s}$, indicating odd photon numbers (blue). **f**, Probability of achieving the target cavity state $|2n+1\rangle_A$, as measured by $P(t)$ for fixed $\Delta\omega_q = -(2n+1)\chi_q$ for cavity A initialized in $|2n\rangle_A$. Error bars reflect the standard error of the mean. These four time-domain curves are fitted using a numerical model of the cascaded pumping process (see Supplementary Fig. 1), resulting in $\Omega = 92 \text{ kHz}$, 88 kHz , 87 kHz and 85 kHz ; and $\lambda = 28 \text{ kHz}$, 27 kHz , 27 kHz and 26 kHz , respectively. The inset shows a block of the cavity process χ matrix for $25 \mu\text{s}$ of PReSPA. The matrix elements $\chi_{nn',n'',n'''}$ are calculated from transmon spectroscopy measurements from all pairs of initial Fock states $|n\rangle_A$ and final Fock states $|n'\rangle_A$.

mixing effects (see Methods), we experimentally obtain $\lambda \approx 27 \text{ kHz}$, $\Omega \approx 88 \text{ kHz}$ for all transition paths.

Characterization of dissipation operator

We characterize the engineered PReSPA operator first by tracking the probability distribution of photon numbers in cavity A over time. This

distribution can be measured using spectroscopy of the ancilla qubit (Fig. 2a), whose frequency shifts by $-\chi_q$ for every additional photon³⁶. As an example, Fig. 2b, c shows dissipative generation of a one-photon Fock state from vacuum under PReSPA as a result of the $|1\rangle\langle 0|$ element of the operator. To demonstrate the simultaneous action of the four parallel conversion paths, we apply PReSPA to an even-parity cat state ($\alpha = 1.6$). As expected, the cavity state converges to an odd-parity state,

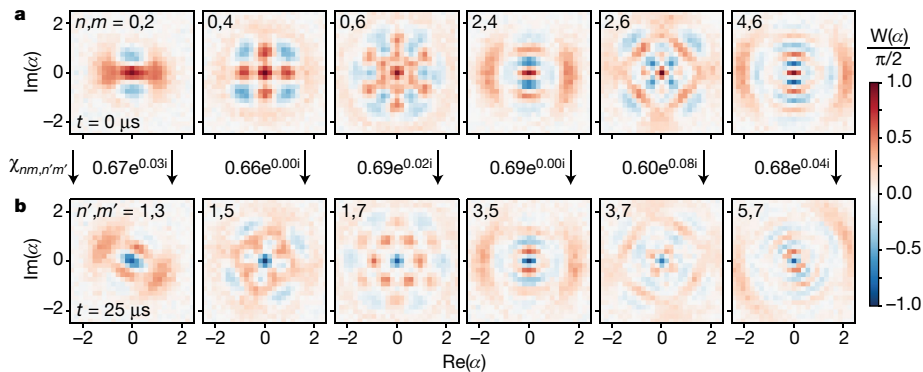


Fig. 3 | Characterization of PReSPA operator: preservation of coherence. **a**, Cavity Wigner tomography of six even-parity superposition states, $(|n\rangle_A + |m\rangle_A)/\sqrt{2}$, prepared by optimal control theory pulses, as input states for PReSPA. **b**, Wigner tomography of the six corresponding output states after $25 \mu\text{s}$ of PReSPA, which are converted approximately to odd-parity superpositions $(|n'\rangle_A + |m'\rangle_A)/\sqrt{2}$, with $n' = n + 1$, $m' = m + 1$. The Wigner function, $W(\alpha)$, a quasi-probability distribution in the oscillator phase space, is directly measured via photon number parity measurements after variable cavity displacements³⁷. From each Wigner function we reconstruct the density matrix, and the most important off-diagonal element is ρ_{nm} (or $\rho_{n'm'}$), which

reflects the coherence between $|n\rangle$ and $|m\rangle$ (or between $|n'\rangle$ and $|m'\rangle$). We also perform similar measurements with permutations of odd-parity superpositions as input states (not shown). The χ matrix block describing the coherence of the process can be computed by combining all the off-diagonal elements in these reconstructed density matrices (Extended Data Fig. 7). The result for the six key elements characterizing PReSPA coherence, $\chi_{nm,n'm'}$, is shown next to the vertical arrows and to a good approximation is equal to $\rho_{n'm'}/\rho_{nm}$. The deviations of $\chi_{nm,n'm'}$ from unity reflects the infidelity of the PReSPA process.

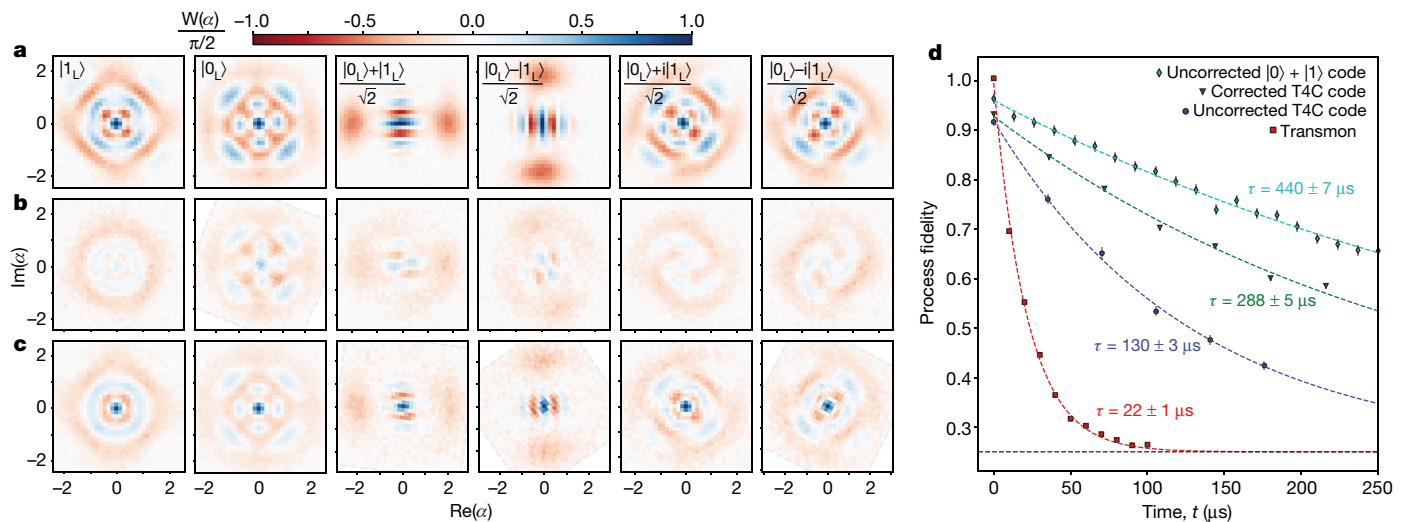


Fig. 4 | AQEC performance. **a–c**, Cavity Wigner tomography for the six cardinal-point states of our T4C code. We measure the states at $t=0$ (**a**), after 143 μs of free evolution (**b**) and after 143 μs of AQEC using PReSPA (**c**). The centre points of the Wigner function show a measurement of state parity, which is well preserved under PReSPA. The Wigner functions have been manually rotated into the correct rotating frame for ease of comparison. **d**, Quantum process fidelity of information storage with the AQEC protocol (green) compared with other reference methods performed with the same

physical system. We use each method to store quantum states corresponding to the six cardinal points of the Bloch sphere, wait for a variable time t (or apply PReSPA for the AQEC curve), retrieve the state via the transmon using a decoding unitary, and perform quantum state tomography of the transmon. Process fidelity is calculated from the quantum state fidelity for the corresponding six measurements. All curves are fitted to the model $\mathcal{F}_{\text{process}}(t) = 0.25 + Ae^{-t/\tau}$ to extract characteristic times and error bars are one standard error.

with all four Fock components shifted up by one photon (Fig. 2d, e). The characteristic timescale of each conversion process can be measured by preparing the Fock state $|2n\rangle$ and tracking the population of $|2n+1\rangle$ as a function of time (Fig. 2f). The four conversion paths show well matched temporal profiles with convergence half-time of $T_{\text{even-odd}} \approx 8 \mu\text{s}$. The fidelity of the conversion decreases for higher photon numbers, consistent with faster photon loss.

To fully benchmark an engineered jump operator like PReSPA, a different type of tomographic characterization and experimental fidelity measure has to be devised. Here for simplicity, we probe the process χ matrix over 25 μs ($\gg T_{\text{even-odd}}$) of PReSPA pumping as a proxy with which to experimentally benchmark the operator. The population conversion from $|n\rangle_A$ to $|n'\rangle_A$ is expressed in the $\chi_{nn',n'',n'''}$ matrix elements, which are measured using spectroscopy and shown in the inset to Fig. 2f.

Quantum coherence of the even-to-odd conversion is reflected in the six key elements of the process matrix, $\chi_{02,13}$, $\chi_{24,35}$, $\chi_{46,57}$, $\chi_{04,15}$, $\chi_{26,37}$ and $\chi_{06,17}$, which can be measured from the transformation of off-diagonal elements of the relevant density matrices. We prepare the six pairwise superposition states of $|0\rangle$, $|2\rangle$, $|4\rangle$ and $|6\rangle$ as input states, and measure the odd-parity output states at $t = 25 \mu\text{s}$ using Wigner tomography³⁷ (Fig. 3). The interference fringes in the output Wigner functions demonstrate the persistence of coherence after photon addition. From density matrix reconstruction, we find that PReSPA converts even-parity superposition states with nearly zero additional phase and an average fidelity of approximately 67%.

For the process χ matrix to reflect the full extent of the parity recovery, we choose a relatively long evolution time of 25 μs so that the initial even-to-odd conversion process is expected to complete with $>98\%$ probability. Various imperfections in the system also accumulate over time, including cavity dephasing, thermal excitations and ongoing single-photon losses that trigger new rounds of PReSPA operation. Therefore, the above characterization should be interpreted as measuring the combined process of a one-time PReSPA operation with a finite success probability together with continuous degradation of quantum states over 25 μs .

AQEC performance

We demonstrate AQEC by applying PReSPA to a set of six cavity multiphoton states corresponding to the cardinal points of the logical Bloch sphere. Over 143 μs we measure the Wigner function of the cavity states, demonstrating that PReSPA preserves the Wigner negativity (a measure of non-classicality) to a greater extent than after uncorrected free evolution (Fig. 4a–c).

We can further retrieve the logical state from the cavity after a variable time with a unitary operation mapping it to the ancilla transmon, $|\psi_L\rangle \rightarrow x|g\rangle_q + y|e\rangle_q$. We maximize this information retrieval by mapping both the logical code word, $|0_L\rangle$, as well as its orthogonal vector, $C_5|1\rangle - C_1|5\rangle$, to the same transmon state $|g\rangle_q$ (likewise for $|1_L\rangle$ and $|e\rangle_q$) to create a decoding that is insensitive to the size of the encoding. We observe a longitudinal relaxation time of $T_1^{-1} = 365 \pm 8 \mu\text{s}$ for the pole states ($|0_L\rangle$ and $|1_L\rangle$) and a transverse relaxation time of $T_2^{-1} = 258 \pm 6 \mu\text{s}$ for the equator states. From the process fidelity of this information storage process (compared to the identity matrix), we determine the corrected logical qubit's lifetime to be $288 \pm 5 \mu\text{s}$, more than twice its uncorrected counterpart (Fig. 4d). However, as in any QEC scheme, redundant storage of information incurs an overhead in decoherence, which in our case causes the uncorrected T4C qubit to decay over three times faster than the uncorrectable bare-cavity qubit (represented by the lowest-energy Fock states). Therefore our corrected qubit's lifetime has not yet reached the QEC break-even point as in a landmark demonstration earlier using digital feedback⁵.

The performance of the current AQEC demonstration is limited by a number of experimental factors, and has substantial room for improvement. (See Supplementary Information for detailed discussions of error channels.) Our ancilla transmon has a spurious thermal excitation rate of 1.4 ms^{-1} , which increases up to 2.0 ms^{-1} during AQEC owing to a combination of heating effect and intrinsic off-resonance excitations. Spontaneous transmon excitation can activate a sequential two-photon-gain process under PReSPA, leading to an incoherent logical bit-flip error (Extended Data Fig. 6). This effect accounts for about 80% of the observed T_1 and about 50% of the T_2 of the logical qubit, which

can be greatly reduced^{38,39} with state-of-the-art filtering and shielding of the thermal environment.

The rest of the logical qubit decoherence can be mostly attributed to the finite success probabilities (S_l and S_t) of PReSPA in correcting single-photon loss. We define $S_i = 1 - G_i^{-1}$ (where subscript i indicates longitudinal or transverse), where G_i is the QEC gain factor between the natural photon-loss rate ($\bar{n}/T_{1A} \approx 1/150 \mu\text{s}^{-1}$) and the part of the corrected logical qubit's error rate (Γ_l or Γ_t) that can still be attributed to photon loss. Various imperfections can each be analysed in terms of a correction failure rate that reduces S_l and S_t . We estimate $S_l = 89\%$ and $S_t = 76\%$, with prominent failure modes including a second photon loss or ancilla decay during a PReSPA operation, virtually activated pumping processes and higher-order cavity nonlinearity (Extended Data Table 3). The QEC success probability should be understood as a hypothetical measure for a standalone PReSPA operation. Actual fidelity of a quantum state that experiences an error-and-correct cycle over a realistic time frame includes additional contributions from new errors at total rates of Γ_l and Γ_t after the initial error is corrected. For example, a logical equator state incurring a photon loss can be expected to recover after $t = 25 \mu\text{s}$ with fidelity $\mathcal{F} \approx S_t e^{-\Gamma_t t} = 69\%$. This is consistent with the characterized χ matrix for PReSPA process, and is further confirmed in a separate experiment using initialized error states (Supplementary Information). The required QEC success probability to reach break-even in the absence of any undesirable transmon excitation is $\frac{1}{3}S_l + \frac{2}{3}S_t > 1 - \frac{2}{3}\bar{n}^{-1} = 81\%$. Modest improvement in ancilla and cavity lifetimes can lift S_l, S_t sufficiently above this threshold to enable AQEC above break-even under realistic experimental conditions (see Methods for simulation results).

Outlook

The four-component cat code (and its truncated variants)^{5,6,17,18} has been pursued as a hardware-efficient paradigm for universal quantum computation³², offering a full gate set and first-order error protection for logical qubits encoded in single cavities. However, there has been a caveat: the repetitive parity checks required to correct single-photon loss^{5,6} are not simultaneously compatible with the continuous driven dissipation needed for phase-space stabilization^{16,25}. A competing bosonic QEC approach based on two-component cat qubits with strongly biased noise channels^{26,40} defers the challenge of photon-loss correction to a next-level repetition code (for example, with a chain of cavities), but comes at the cost of increased hardware complexity.

Our introduction of PReSPA provides a non-invasive method for photon-loss correction, paving the road for concurrent logic operations or dissipative stabilization within the odd-parity subspace. For example, modest four-photon dissipation^{16,32} induced by a second reservoir can be applied together with PReSPA to evacuate entropy from cat-size changes and correct for phase drifts, hence completing fully passive first-order protection of a single-cavity logical qubit. Autonomously corrected quantum gates can be constructed from path-independent unitary operations acting on the code space and error space in parallel^{17,18,41}. To improve fault tolerance, forward propagation of ancilla relaxation errors can be suppressed by error-transparent processes through the second excited state of the transmon¹⁷ or by using a biased-noise ancilla^{40,42}. In addition, our cascaded pumping technique to realize path erasure can be generalized to construct a broad family of dissipation operators of the form $\hat{L} = \sum_{i,j} \lambda_{i,j} |i\rangle\langle j|$ in the Fock basis, providing the tools for realizing various other AQEC schemes^{11,13,20}.

For reducing errors in quantum computing, the development of intrinsically protected physical qubits by Hamiltonian engineering^{43,44} and implementation of QEC codes based on redundancy^{1,19} have often been considered two distinct pursuits. The establishment of continuous AQEC of prominent errors, joined by recent developments of

driven qubits with biased noise channels^{26,40}, bridges this divide. In our demonstration, the cavity qubit is protected by a QEC code executed through a driven-dissipative environment in quasi-equilibrium, as described by a time-independent rotating-frame Hamiltonian and dissipation operators. This feature of passive correction is similar to the scenario envisioned as self-corrected quantum memory⁴⁵, but is realized by adding corrective dissipation instead of increasing the energy barrier against error processes. Beyond specific implementations in circuit QED, our work suggests reservoir engineering as a unifying force that applies the flexibility of code-based error correction to improve the robustness of physical qubits with minimal resource overhead.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-021-03257-0>.

- Lidar, D. A. & Brun, T. A. (eds) *Quantum Error Correction* 1st edn (Cambridge Univ. Press, 2013).
- Schindler, P. et al. Experimental repetitive quantum error correction. *Science* **332**, 1059–1061 (2011).
- Cramer, J. et al. Repeated quantum error correction on a continuously encoded qubit by real-time feedback. *Nat. Commun.* **7**, 11526 (2016).
- Kelly, J. et al. State preservation by repetitive error detection in a superconducting quantum circuit. *Nature* **519**, 66–69 (2015).
- Ofek, N. et al. Extending the lifetime of a quantum bit with error correction in superconducting circuits. *Nature* **536**, 441–445 (2016).
- Hu, L. et al. Quantum error correction and universal gate set operation on a binomial bosonic logical qubit. *Nat. Phys.* **15**, 503–508 (2019).
- Andersen, C. K. et al. Repeated quantum error detection in a surface code. *Nat. Phys.* **16**, 875–880 (2020).
- Ahn, C., Doherty, A. C. & Landahl, A. J. Continuous quantum error correction via quantum feedback control. *Phys. Rev. A* **65**, 042301 (2002).
- Atalaya, J. et al. Continuous quantum error correction for evolution under time-dependent Hamiltonians. Preprint at <https://arxiv.org/abs/2003.11248> (2020).
- Kerckhoff, J., Nurdin, H. I., Pavlichin, D. S. & Mabuchi, H. Designing quantum memories with embedded control: photonic circuits for autonomous quantum error correction. *Phys. Rev. Lett.* **105**, 040502 (2010).
- Kapit, E. Hardware-efficient and fully autonomous quantum error correction in superconducting circuits. *Phys. Rev. Lett.* **116**, 150501 (2016).
- Reiter, F., Sørensen, A. S., Zoller, P. & Muschik, C. A. Dissipative quantum error correction and application to quantum sensing with trapped ions. *Nat. Commun.* **8**, 1822 (2017).
- Albert, V. V. et al. Pair-cat codes: autonomous error-correction with low-order nonlinearity. *Quant. Sci. Technol.* **4**, 035007 (2019).
- Sarovar, M. & Milburn, G. J. Continuous quantum error correction by cooling. *Phys. Rev. A* **72**, 012306 (2005).
- Brune, M. et al. Observing the progressive decoherence of the “meter” in a quantum measurement. *Phys. Rev. Lett.* **77**, 4887–4890 (1996).
- Mundhada, S. et al. Experimental implementation of a Raman-assisted eight-wave mixing process. *Phys. Rev. Appl.* **12**, 054051 (2019).
- Reinhold, P. et al. Error-corrected gates on an encoded qubit. *Nat. Phys.* **16**, 822–826 (2020).
- Ma, Y. et al. Error-transparent operations on a logical qubit protected by quantum error correction. *Nat. Phys.* **16**, 827–831 (2020).
- Knill, E. & Laflamme, R. Theory of quantum error-correcting codes. *Phys. Rev. A* **55**, 900–911 (1997).
- Lihm, J.-M., Noh, K. & Fischer, U. R. Implementation-independent sufficient condition of the Knill-Laflamme type for the autonomous protection of logical qubits by strong engineered dissipation. *Phys. Rev. A* **98**, 012317 (2018).
- Poyatos, J. F., Cirac, J. I. & Zoller, P. Quantum reservoir engineering with laser cooled trapped ions. *Phys. Rev. Lett.* **77**, 4728–4731 (1996).
- Krauter, H. et al. Entanglement generated by dissipation and steady state entanglement of two macroscopic objects. *Phys. Rev. Lett.* **107**, 080503 (2011).
- Barreiro, J. T. et al. An open-system quantum simulator with trapped ions. *Nature* **470**, 486–491 (2011).
- Kienzler, D. et al. Quantum harmonic oscillator state synthesis by reservoir engineering. *Science* **347**, 53–56 (2015).
- Leghtas, Z. et al. Confining the state of light to a quantum manifold by engineered two-photon loss. *Science* **347**, 853–857 (2015).
- Lescanne, R. et al. Exponential suppression of bit-flips in a qubit encoded in an oscillator. *Nat. Phys.* **16**, 509–513 (2020).
- Koch, J. et al. Charge-insensitive qubit design derived from the Cooper pair box. *Phys. Rev. A* **76**, 042319 (2007).

28. Reagor, M. et al. A quantum memory with near-millisecond coherence in circuit QED. *Phys. Rev. B* **94**, 014506 (2016).
29. Axline, C. et al. An architecture for integrating planar and 3D cQED devices. *Appl. Phys. Lett.* **109**, 042601 (2016).
30. Chuang, I. L., Leung, D. W. & Yamamoto, Y. Bosonic quantum codes for amplitude damping. *Phys. Rev. A* **56**, 1114–1125 (1997).
31. Terhal, B. M., Conrad, J. & Vuillot, C. Towards scalable bosonic quantum error correction. *Quant.Sci. Technol.* **5**, 043001 (2020).
32. Mirrahimi, M. et al. Dynamically protected cat-qubits: a new paradigm for universal quantum computation. *New J. Phys.* **16**, 045014 (2014).
33. Cohen, J., Smith, W. C., Devoret, M. H. & Mirrahimi, M. Degeneracy-preserving quantum nondemolition measurement of parity-type observables for cat qubits. *Phys. Rev. Lett.* **119**, 060503 (2017).
34. Pinotsi, D. & Imamoglu, A. Single photon absorption by a single quantum emitter. *Phys. Rev. Lett.* **100**, 093603 (2008).
35. Macklin, C. et al. A near-quantum-limited Josephson traveling-wave parametric amplifier. *Science* **350**, 307–310 (2015).
36. Schuster, D. I. et al. Resolving photon number states in a superconducting circuit. *Nature* **445**, 515–518 (2007).
37. Vlastakis, B. et al. Deterministically encoding quantum information using 100-photon Schrödinger cat states. *Science* **342**, 607–610 (2013).
38. Serniak, K. et al. Direct dispersive monitoring of charge parity in offset-charge-sensitive transmons. *Phys. Rev. Appl.* **12**, 014052 (2019).
39. Jin, X. Y. et al. Thermal and residual excited-state population in a 3D transmon qubit. *Phys. Rev. Lett.* **114**, 240501 (2015).
40. Grimm, A. et al. Stabilization and operation of a Kerr-cat qubit. *Nature* **584**, 205–209 (2020).
41. Ma, W.-L. et al. Path-independent quantum gates with noisy ancilla. *Phys. Rev. Lett.* **125**, 110503 (2020).
42. Puri, S. et al. Stabilized cat in a driven nonlinear cavity: a fault-tolerant error syndrome detector. *Phys. Rev. X* **9**, 041009 (2019).
43. Douçot, B. & Ioffe, L. B. Physical implementation of protected qubits. *Rep. Prog. Phys.* **75**, 072001 (2012).
44. Gyenis, A. et al. Experimental realization of an intrinsically error-protected superconducting qubit. Preprint at <https://arxiv.org/abs/1910.07542> (2019).
45. Brown, B. J., Loss, D., Pachos, J. K., Self, C. N. & Wootton, J. R. Quantum memories at finite temperature. *Rev. Mod. Phys.* **88**, 045005 (2016).
46. Heeres, R. W. et al. Implementing a universal gate set on a logical qubit encoded in an oscillator. *Nat. Commun.* **8**, 94 (2017).

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Methods

Approximate AQEC and decoding unitary

The T4C code belongs to the class of bosonic code with $N=2$ rotational symmetry⁴⁷. It can be understood as an approximate version of a Schrödinger cat code^{32,48} at small photon numbers. In the idealized limit of large photon numbers and instantaneous operations of $\hat{\Pi}' = \sum_{n=0}^{\infty} |2n+1\rangle\langle 2n|$ upon photon loss, such a cat code would enable perfect retrieval of quantum information. Here we have extended the PReSPA operator to an infinite series $\hat{\Pi}'$, which can be viewed as a parity-selective version of the Susskind–Glogower bare raising operator⁴⁹. Practical limitations, however, necessitate a compromise accounting for the growth of the photon loss rate with average photon number and the increasing technical challenge of expanding the terms in the dissipation operator. As a compromise, we choose the T4C code containing the code words in equation (1), composed of odd-numbered Fock states $|n\rangle$ with $n \leq 7$.

Within T4C encoding, the joint effect of loss and instantaneous PReSPA, as described by a combined loss operator $\hat{\Pi}'\hat{a}$, generally induces an unwanted distortion of the original quantum state. Specifically, given a number of j loss/PReSPA events within a time period t , the initial state $|\psi_0\rangle = x|0_L\rangle + y|1_L\rangle$ evolves into the state

$$|\psi_j(t)\rangle = x n_j^{15}(t) |\psi_j^{15}(t)\rangle + y n_j^{37}(t) |\psi_j^{37}(t)\rangle. \quad (5)$$

Here, $|\psi_j^{kl}(t)\rangle = \|\sum_{n=k,l} C_n n^{j/2} \exp(-nt/2T_{1A}) |n\rangle\| / [n_j^{kl} N_j(t)]$ are orthonormal, and the coefficients recording the unwanted state distortion are $n_j^{kl}(t) = \|\sum_{n=k,l} C_n n^{j/2} \exp(-nt/2T_{1A}) |n\rangle\| / N_j(t)$ where $k, l=1, 5$ or $3, 7$.

The decoding transformation seeks to transfer the amplitudes x, y from the logical cavity qubit to the auxiliary transmon. Perfect transfer is impeded by two key factors. First, the coefficients $n_j^{kl}(t)$ in equation (5) will generally differ and, hence, lead to the mentioned distortion of x and y . Second, the code words undergo rotations within the $\mathcal{H}_{15}$ and $\mathcal{H}_{37}$ subspaces described by

$$|\psi_j(t)\rangle = x n_j^{15}(t) (\cos\theta_j |u_0\rangle + \sin\theta_j |u_1\rangle) + y n_j^{37}(t) (\cos\varphi_j |v_0\rangle + \sin\varphi_j |v_1\rangle). \quad (6)$$

Here, the angles θ_j and φ_j depend both on time and the number of loss events. See Extended Data Fig. 1a for a visual description of the effect of jump and no-jump dynamics. The states $\{|u_0\rangle, |u_1\rangle\}$ and $\{|v_0\rangle, |v_1\rangle\}$ denote orthonormal bases of the $\mathcal{H}_{15}$ and $\mathcal{H}_{37}$ subspaces, respectively.

Constructing a decoding unitary $\hat{U}_{\text{drive}}$ such that:

$$\begin{aligned} |g\rangle \otimes |u_0\rangle &\mapsto |g0\rangle, & |g\rangle \otimes |u_1\rangle &\mapsto |g1\rangle, \\ |g\rangle \otimes |v_0\rangle &\mapsto |e0\rangle, & |g\rangle \otimes |v_1\rangle &\mapsto |e1\rangle, \end{aligned} \quad (7)$$

one obtains the reduced transmon density matrix

$$\hat{\rho}_q = \sum_j p_j \begin{pmatrix} |x|^2 (n_j^{15})^2 & x^* y n_j^{15} n_j^{37} \cos\zeta_j \\ xy^* n_j^{15} n_j^{37} \cos\zeta_j & |y|^2 (n_j^{37})^2 \end{pmatrix}, \quad (8)$$

with p_j denoting the probability of observing j loss events, and $\zeta_j = \theta_j - \varphi_j$. In general, θ_j and φ_j will not be identical, thus causing a second contribution to amplitude distortions. Nevertheless, they vary in the same direction as j varies, and this partial cancellation plays an important part in limiting the intrinsic loss of information in this approximate AQEC protocol. See Supplementary Information section I for analytic details.

We further note that the photon-number expectation value, when averaged over all trajectories, remains exactly constant under the effect of $\hat{\Pi}'\hat{a}$. In essence, despite its continuous decrease (under no-jump evolution) and stochastic increase (due to PReSPA after a photon loss), the effective cat size $|a|$ of the encoding remains a constant on average. This is different from the cat states in parity-measurement-based

bosonic QEC experiments^{5,17}. In these experiments, the cat states decrease deterministically in size over a timescale of $1/2\kappa$, hence quickly experiencing wavefunction overlap between C_a and C_{ia} . The T4C code thus has the advantage of being self-sustained in energy, and the loss of orthogonality at small α is only encountered through a (2nd-order) diffusive process.

We quantify deviations of equation (8) from the intended target transmon states with the process fidelity $\mathcal{F}$ obtained by averaging $\mathcal{F}_{xy}(t) = |\langle \psi_q | \hat{\rho}_q(t) | \psi_q \rangle|^2$ over the six cardinal Bloch sphere points and rescaling its range to $1/4 \leq \mathcal{F} \leq 1$. Extended Data Fig. 1b compares the theoretical process fidelity for PReSPA-corrected T4C (using the code-word coefficients in our experiment as shown below equation (1)) to free evolution without PReSPA, for both T4C and Fock-state encoding. Theoretically, the corrected logical qubit incurs no longitudinal relaxation, and the transverse relaxation rate Γ_t is 40 times lower than the single-photon loss rate $\bar{n}\kappa$, corresponding to an intrinsic QEC failure rate of $1 - S_t = 2.5\%$.

The T4C code can also be viewed as an instance of the binomial code class⁵⁰, assuming the two logical codes contain equal photon numbers ($\bar{n}$). Indeed, informational leakage associated with photon loss is minimized with equal $\bar{n}$ for $|0_L\rangle$ and $|1_L\rangle$, and the slight imbalance in our experimental code ($\bar{n} = 3.6, 3.4$ for $|0_L\rangle, |1_L\rangle$) was incidental and not optimal. However, we calculated that a revision to an equal $\bar{n}$ of 3.5 would only improve S_t by 0.08%, which is extremely marginal compared with all the imperfections that lead to $1 - S_t \approx 24\%$ in the experiment. Assuming balanced $\bar{n}$ in a binomial code, the value of $\bar{n}$ can be chosen to balance the trade-off between the intrinsic QEC failure rate (minimized at $\bar{n} = 4$) and photon-loss rate (minimized at $\bar{n} = 3$, the smallest possible in this code). Given the coherence parameters of the transmon and the cavity in this present study, the AQEC performance can be improved by choosing a smaller $\bar{n}$ close to 3.

Device and fabrication

Our device uses a 3D-planar-hybrid circuit quantum electrodynamics (QED) architecture²⁹, and the design specifics are similar to those in ref. ⁵¹. The cavity is machined from 5N5 Al (99.9995% pure) and houses a large waveguide section containing two cylindrical re-entrant quarter-wave resonators²⁸ and a small waveguide tunnel to fit a sapphire chip. The transmon and the low- Q (quality factor) readout resonator are made from thin-film aluminium deposited on the sapphire chip. The transmon contains a single Al-AlO_x-Al Josephson junction and is fabricated using a Dolan bridge technique. Electron-beam lithography is carried out with a 30-keV JEOL JSM-7001F SEM, and the evaporation/oxidation is performed with a Plassys MEB550S evaporator.

Of the two high- Q storage cavity modes, the lower-frequency mode A is used for this experiment while the higher-frequency (6.089 GHz) mode, B, is idle in the vacuum state throughout the experiment. Cavity mode B has a dispersive coupling with the transmon of 6.5 MHz that we found too high for implementing PReSPA. It has a thermal population of 1–2%, which has a small contribution to the dephasing of mode A. All relevant device parameters are listed in Extended Data Table 1.

Driven Josephson circuit Hamiltonian under PReSPA

The Hamiltonian of the circuit QED system can be derived using the perturbation theory⁵² that adds weak anharmonicity to three harmonic oscillator modes corresponding to the cavity A, transmon ancilla q and the stripline resonator R:

$$\frac{\hat{H}}{\hbar} = \tilde{\omega}_q \hat{q}^\dagger \hat{q} + \tilde{\omega}_A \hat{a}^\dagger \hat{a} + \tilde{\omega}_R \hat{r}^\dagger \hat{r} - \frac{E_J}{\hbar} \left(\cos\hat{\varphi} + \frac{\hat{\varphi}^2}{2} \right) \quad (9)$$

where $\tilde{\omega}_q, \tilde{\omega}_A, \tilde{\omega}_R$ are the frequencies of the eigenmodes of the linearized system, $\hat{q}, \hat{a}$ and $\hat{r}$ are their lowering operators, E_J is the Josephson energy of the junction, and $\hat{\varphi} = \phi_q \hat{q} + \phi_A \hat{a} + \phi_R \hat{r} + \text{h.c.}$ is the phase operator of

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the Josephson junction. ϕ_q, ϕ_A, ϕ_R are the mode zero-point phase fluctuations across the junction for modes q, A and R.

Implementing PReSPA requires two frequency combs: one comb exciting the transmon, and one mixing comb converting that transmon excitation into excitations in A and R. This mixing comb activates a four-wave mixing process using the nonlinearity of the Josephson junction. To account for the four mixing tones with frequencies ω_m in our Hamiltonian, we make a unitary transformation by displacing the qubit annihilation operator $\hat{q} \rightarrow \hat{q} + \sum_m \xi_m e^{i\omega_m t}$, and the phase across the junction is (see, for example, the Supplementary Information of ref. 25):

$$\hat{\phi} = \phi_q \hat{q} + \phi_A \hat{a} + \phi_R \hat{r} + \sum_{m=0}^3 \phi_q \xi_m e^{i\omega_m t} + \text{h.c.} \quad (10)$$

Expanding the cosine, going to a rotating frame, and employing a dispersive transformation that cancels 2nd-order terms, the relevant 4th- and 6th-order terms include non-driven terms (without ξ_m) and driven terms. The non-driven terms, generic to most circuit QED systems, are:

$$\frac{\hat{H}_{\text{non-driven}}}{\hbar} = -\chi_q \hat{q}^\dagger \hat{q} \hat{a}^\dagger \hat{a} - \chi_R \hat{q}^\dagger \hat{q} \hat{r}^\dagger \hat{r} - \chi_{AR} \hat{a}^\dagger \hat{a} \hat{r}^\dagger \hat{r} - \frac{K}{2} \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} - \frac{\chi'_q}{2} \hat{q}^\dagger \hat{q} \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} \quad (11)$$

where $\hat{H}_{\text{non-driven}}$ are the non-driven terms, $\chi_q, \chi_R, \chi_{AR}$ are the dispersive couplings between the transmon and cavity A, between the transmon and reservoir R, and between A and R, respectively. K is the self Kerr of cavity A, and χ'_q is the 6th-order nonlinearity. We treat the transmon as a two-level system because the higher excited states are not accessed in this experiment. Likewise, for all operations except readout, R is either in the ground or first excited state and we ignore its higher order terms.

By setting $\omega_m = \omega_A + \omega_R - \omega_q + m\eta$ we get stationary, or slowly rotating, four-wave mixing terms:

$$\frac{\hat{H}_{\text{mix}}}{\hbar} = -\frac{E_J}{\hbar} \phi_q^2 \phi_A \phi_R \sum_{m=0}^3 \xi_m e^{im\eta t} \hat{q} \hat{a}^\dagger \hat{r}^\dagger + \text{h.c.} \quad (12)$$

where η is the difference in frequency between each nearest pair of mixing tones. The four mixing tones will each individually Stark shift the transmon but we can simply absorb those into the transmon frequency for the rotating frame transformation mentioned above. There will also be slowly rotating Stark shift terms that are a result of the cross terms of two different mixing tones:

$$\frac{\hat{H}_{\text{Stark}}}{\hbar} = -\frac{E_J}{\hbar} \phi_q^4 \sum_{k=0}^3 \sum_{l=0}^3 \xi_k \xi_l^* e^{-i(l-k)\eta t} \hat{q}^\dagger \hat{q} \quad (13)$$

The four mixing tones acting together drive the four $|2n, e, 0\rangle \leftrightarrow |2n+1, g, 1\rangle$ ($n = 0, 1, 2, 3$) transitions (the black solid arrows in Fig. 1c). We can calculate the complex Rabi rate of these driven transitions under a weak-drive approximation of $|\xi_k|^2 \ll \hbar\eta/(E_J\phi_q^4)$:

$$\Omega_n = -\frac{E_J}{\hbar} \sqrt{2n+1} \phi_q^2 \phi_A \phi_R \times \left[\xi_n - \frac{E_J \phi_q^4}{\hbar\eta} \sum_{m; k; l \neq k} \frac{\xi_m}{(l-k)} (\xi_k \xi_l^* \delta_{n-m, l-k} - \xi_k^* \xi_l \delta_{n-m, k-l}) + \mathcal{O}\left(\frac{E_J^2 \phi_q^8 \xi^4}{\hbar^2 \eta^2}\right) \right] \quad (14)$$

where δ is the Kronecker delta function. This rate is caused by two different mechanisms: a direct drive by one of the four tones that is on resonance, and a multi-tone parametric effect. This parametric effect arises from the time-modulation of transmon frequency by the Stark shift induced by pairs of mixing tones. When the detuning of one of the off-resonant mixing tones exactly matches the modulation frequency, the transition can be parametrically driven by a combination of three tones. Depending on the relative phases of the three tones, these terms can contribute constructively or destructively to the transition rate. Owing to the bosonic enhancement factor $\sqrt{2n+1}$, higher photon number states typically require smaller ξ_n .

This calculation has been done with the assumption that the spacing between tones η is constant, but we could also consider non-even frequency spacing. In this case, terms that rotate very slowly with time would appear in the mixing rates, which would cause our PReSPA operator to change undesirably with time. We make the choice of keeping η constant to avoid these complications.

A similar calculation can be done for the four selective transmon drives. To preserve path independence we use the same frequency-spacing magnitude η as the mixing drives but with the opposite sign. As the transmon drives are relatively weak, they do not cause a substantial single-tone or multi-tone Stark shift but the frequency modulation caused by the mixing drives will allow for similar three-tone parametric mixing effects (two mixing tones and one transmon tone). We can write the rates for the four transmon transitions as:

$$\lambda_n = \Lambda_n - \frac{2E_J \phi_q^4}{\hbar\eta} \sum_{m \neq n; k; l \neq k} \frac{\Lambda_m \xi_k \xi_l}{m-n} \delta_{n-m, k-l} \quad (15)$$

where Λ_n are the bare transmon Rabi rates without the mixing drives. For either comb, the effect of this parametric mixing is a renormalization of the Rabi rates Ω_n and λ_n from the Rabi rates by individual resonant tones (that are proportional to ξ_n and Λ_n).

PReSPA spectroscopy and rates

Implementing PReSPA requires initial parameter guesses supplemented with experimental corrections. We make the initial assumption that the comb frequency spacing $\eta = 2\chi_q$ and that the $|0e0\rangle \leftrightarrow |1g1\rangle$ (mixing) transition frequency is $\omega_A + \omega_R - \omega_q$. Because our strong off-resonant drives Stark shift the transmon, a detuning Δ needs to be added to the mixing comb frequencies and subtracted from the transmon comb frequencies to best match the desired transitions.

In addition, the presence of non-zero χ'_q adds a quadratic shift to the transmon frequency based on the cavity photon occupation. Therefore, there is no choice of evenly spaced tones that can drive all transitions on resonance. To best match the experimentally measured transitions we calibrate η and Δ through a set of spectroscopy measurements (Extended Data Fig. 2c). Experimentally, we choose $\eta = 2.679$ MHz and $\Delta \approx 2.9$ MHz (calibrated on a daily basis) to ensure no pair of tones are farther than 10 kHz (about $2\chi'_q$) off-resonant to their corresponding transitions.

To match the dissipative processes across the four parallel paths in PReSPA, we measure the probability of photon addition, over time, to each of the four even states (see Supplementary Fig. 1 for further details). By fitting the curves of photon population and transmon excitation probability we can extract Ω_n and λ_n for each of the four conversion paths.

From equations (14) and (15) we can also estimate Ω_n and λ_n from the amplitudes and phases of the microwave tones. We find experimentally that achieving equal Ω_n and λ_n for the four conversion paths requires different microwave amplitudes within each comb, in quantitative agreement with theoretical predictions (Extended Data Table 2). There is a discrepancy in a global pre-factor in the magnitude of Ω_n , possibly caused by the coarse estimates of zero point fluctuation parameters

(ϕ values) from measurable experimental parameters (that is, frequencies and dispersive shifts).

Cavity and transmon Ramsey under PReSPA

To characterize the phase evolution of various multiphoton superposition states under PReSPA, we perform a Ramsey-style experiment (PReSPA Ramsey, as shown in Extended Data Fig. 3). This can be used both for measuring the frequency and coherence of odd-parity superposition states under PReSPA and for tuning up superposition phases in the PReSPA operator $\hat{U}$.

We measure the phase imparted in the photon addition process by starting with an equal superposition of two even states (for example, $\frac{|0\rangle + |2\rangle}{\sqrt{2}}$) and perform a PReSPA Ramsey at times greater than 25 μs , sufficient for nearly complete even-to-odd conversion. By fitting the Ramsey oscillations, we can extract the phase of the PReSPA-converted state (for example, $\frac{|1\rangle + |3\rangle}{\sqrt{2}}$), which is calibrated against the phase extracted from another PReSPA Ramsey experiment on the corresponding odd-parity initial state (for example, $\frac{|1\rangle + |3\rangle}{\sqrt{2}}$). Phase and amplitude parameters of the microwave combs are adjusted to minimize the imparted phases of the photon addition process.

PReSPA Ramsey is further used to measure the phase accumulation rate of our logical code words. When measuring the process fidelity of error correction we work in a frame where $|0\rangle_L$ is stationary. By measuring the phase of $|1\rangle_L$ as a function of time we can calculate the correct decoding angle.

We also perform a regular qubit Ramsey experiment for the transmon in the presence of the mixing combs of PReSPA (without exciting cavity A). This allows measurement of the Stark shift of the transmon rapidly and accurately, because to a very good approximation, only the mixing comb contributes to the Stark shift of the transmon. This experiment further shows that the coherence time of the transmon is not affected by the strong off-resonant drives of PReSPA (with $T_{2q}^* = 17 \mu\text{s}$).

GRAPE methods

Pulses for state preparation and decoding are constructed using quantum optimal control (QOC) theory^{46,53,54,55}, as adapted for arbitrary cavity-state controls in circuit QED in ref.⁴⁶ (see Supplementary Information section 2 for additional details). QOC numerically optimizes the envelopes of tones acting on the transmon and storage cavity, such that deviations of the realized state or unitary from the target state or target unitary are minimized. Deviations are quantified as state or process infidelities and incorporated into a cost functional C , which is subject to gradient-descent minimization via gradient ascent pulse engineering (GRAPE)⁵⁶.

Two envelope-modulated drives with carrier frequencies ω_A and ω_q are applied to storage cavity and transmon for preparation of the cavity states shown in Figs. 3a and 4a. By adjusting the respective envelopes, GRAPE maximizes the state-transfer fidelity $\mathcal{F}_{\text{prep}} = |\langle\psi_T|\psi\rangle|^2$, where $|\psi_T\rangle$ and $|\psi\rangle$ are the cavity target state and the actual cavity state realized by the pulse. Closed-system time evolution is simulated based on the rotating-frame Hamiltonian (11).

GRAPE is further used in constructing a decoding unitary $\hat{U}_{\text{drive}}$ seeking to map the T4C-encoded cavity state to the transmon. As minimal constraints on $\hat{U}_{\text{drive}}$ it is required to act on the four relevant basis states as shown in equation (7). This does not fully determine the unitary in the seven-dimensional subspace $\mathcal{S}$ spanned by $|g, n\rangle$ and $|e, \ell\rangle$ where $n = 0, 1, 3, 5, 7$ and $\ell = 0, 1$. We therefore specified three additional constraints for the numerical optimization. However aside from preserving unitarity, these additional constraints were discretionary. Beyond unitarity, no constraints were placed in the algorithm on the action of $\hat{U}_{\text{drive}}$ on states in the remaining space orthogonal to $\mathcal{S}$. In this study, the actual form of transformation on those states is not characterized

and may have a small positive or negative impact on the fidelity of decoded qubit state. Future enforcement of additional constraints may lead to small improvements in logical fidelity.

Cavity Kerr effect on decoding

The truncated cat states evolve with a deterministic collapse-revival cycle due to the self-Kerr effect introduced by the Josephson junction⁵⁷. This effect does not directly cause logical qubit decay (except for a small reduction to QEC success probability S_c ; see Extended Data Table 3). However, it imposes an inconvenience that for an arbitrary time t (modulo π/K or 288 μs), a dedicated optimal control theory decoding pulse is needed to account for the phase accumulation due to the self Kerr, K . In our experiment, by accounting for an effective Z-rotation of the logical qubit after decoding, we can recycle the same decoding pulse every $\pi/4K$ or 72 μs .

To see this, in the rotating frame chosen to make $|0\rangle_L$ stationary (calibrated by the PReSPA Ramsey; see Extended Data Fig. 3), the Kerr evolution of a T4C state (without decoherence) is:

$$|\psi\rangle = x(C_1|1\rangle + C_5|5\rangle) + ye^{-2iKt}(C_3|3\rangle + C_7e^{8iKt}|7\rangle) \quad (16)$$

In the experiment, we computed two different decoding pulses. The first decoding pulse, used for $t = 0 \pmod{72 \mu\text{s}}$, implements the original transformation prescribed by equation (7):

$$\begin{aligned} |g\rangle \otimes (C_1|1\rangle + C_5|5\rangle) &\mapsto |g0\rangle, \\ |g\rangle \otimes (C_1|5\rangle - C_5|1\rangle) &\mapsto |g1\rangle, \\ |g\rangle \otimes (C_3|3\rangle + C_7|7\rangle) &\mapsto |e0\rangle, \\ |g\rangle \otimes (C_3|7\rangle - C_7|3\rangle) &\mapsto |e1\rangle, \end{aligned} \quad (17)$$

The second decoding pulse, used for $t = 36 \mu\text{s} \pmod{72 \mu\text{s}}$, implements a modified transformation where the coefficient of $|7\rangle$ acquires an extra minus sign in equation (17).

These two decoding pulses differ slightly in their performance. This is responsible for the alternating pattern in the state and process fidelity of T4C code (as a function of time) in Fig. 4b and Extended Data Fig. 4b, c.

Simulation of experimental AQEC

To verify the error contribution from various experimental imperfections, we perform numerical simulations of the AQEC experiment by solving the master equation in QuTiP. The simulation is carried out in a Hilbert space of $9 \times 2 \times 2$ dimensions and a rotating frame of frequencies $\omega_q, \omega_A, \omega_R$ for the transmon, cavity and reservoir excitations. The fast dynamics on the scale of mode frequencies are therefore not simulated, but the relative dynamics of different photon-number states on the timescale of about χ_q are fully captured.

The simulated system Hamiltonian is composed of the non-driven terms $\hat{H}_{\text{non-drive}}$ (equation (11) with parameters listed in Extended Data Table 1) and the following time-dependent drive Hamiltonian:

$$\begin{aligned} \hat{H}_{\text{drive}}/\hbar &= \sum_{n=0}^3 e^{-i[2n\chi_q + (5n-2)\chi'_q]t} \sum_{m=0}^8 \left(\lambda_n |m, e, 0\rangle \langle m, g, 0| \right. \\ &\quad \left. + \Omega_n \sqrt{\frac{m+1}{2n+1}} |m, e, 0\rangle \langle m+1, g, 1| \right) + \text{h.c.} \end{aligned} \quad (18)$$

This drive Hamiltonian describes four transmon excitation tones and four four-wave mixing tones that are $\pm 2\chi'_q$ detuned from the transition frequencies, as shown in Extended Data Fig. 2. λ_n and Ω_n are the transmon drive rate and four-wave mixing rate for the intended transitions as listed in Extended Data Table 2, accounting for the small differences in rates within each comb. Crucially, equation (18) encapsulates the resonant and off-resonant dynamics of each photon-number state ($m = 0$ to 8) under the effect of all four tones ($n = 0$ to 3) in each PReSPA comb. We assume the constant Stark shift of all modes have been included in

the rotating frame. We also make the assumption that the effect of two-tone modulated Stark shift is fully captured by the renormalization of comb rates λ_n and Ω_n in the spirit of equations (14) and (15).

In addition to single photon loss, we account for the spurious excitations, relaxation, dephasing of the ancilla and other causes of cavity dephasing. The QuTip simulation thus solves the master equation,

$$\begin{aligned} \frac{\partial \rho(t)}{\partial t} = & -\frac{i}{\hbar}[(\hat{H}_{\text{non-drive}} + \hat{H}_{\text{drive}}), \rho(t)] \\ & + \left(\frac{1}{T_{1A}} D[\hat{a}] + \frac{1}{T_{1R}} D[\hat{r}] + \frac{1}{T_{1q}} D[\hat{q}] \right. \\ & \left. + \frac{1}{T_{\phi}} D[\hat{q}^{\dagger} \hat{q}] + \gamma_r D[\hat{q}^{\dagger}] + \gamma_{\phi} D[\hat{a}^{\dagger} \hat{a}] \right) \rho(t) \end{aligned} \quad (19)$$

where $D[\hat{O}]\rho(t) = \hat{O}\rho\hat{O}^{\dagger} - \frac{1}{2}\{\hat{O}^{\dagger}\hat{O}, \rho\}$. The T_{1A} , T_{1R} , T_{1q} are the relaxation times for cavity, reservoir and transmon, respectively. γ_r is the spurious ancilla excitation rate and γ_{ϕ} is the storage cavity dephasing rate.

The density matrix of the system at any given time t is transformed by the decoding unitary equation (7) so that the cavity logical information is decoded into the transmon state just like in the experiment. The cavity Kerr effect is taken into account by adjusting the phases of the decoding unitary as discussed earlier in Methods. The quantum state fidelity against the targeted ancilla state is then computed as a function of time t .

The simulation shows a corrected logical qubit lifetime of 290 μs , with $T_1^{-1} = 383 \mu\text{s}$ and $T_1^{-1} = 255 \mu\text{s}$, in excellent agreement with the experiment. We also performed sensitivity analysis to individual Hamiltonian or loss parameters in the simulation, and the results are consistent with our estimates of individual error channels presented in Extended Data Table 3 and the Supplementary Information section 3.

Since the simulation quantitatively captures all the error channels displayed in the experiment, we can evaluate the performance of our PReSPA pumping scheme when the system parameter improves. With better cavity and transmon coherence, for the same static and driven Hamiltonian parameters, the logical error rates due to transmon decay and the second photon decay will be suppressed quadratically, as expected for an ideal first-order QEC protocol. However, other error channels due to higher-order nonlinearity and drive-tone selectivity will not scale down equally. Therefore, reduced χ_q , κ , Ω and λ should be chosen to partially trade away the gain from coherence improvement to suppress these error channels. This is analogous to the situation in active QEC experiments, where less frequent parity measurements may be desirable when coherence time improves^{5,6}.

Extended Data Fig. 5a demonstrates the simulated AQEC performance including all the imperfections intrinsic to a transmon-based PReSPA scheme, which is computed using a specific set of Hamiltonian parameters (corresponding to 50% lower PReSPA rates and a 80% lower dispersive shift than the experiment). The result shows logical lifetime 7–9 times longer than the physical photon loss time for a $\bar{n} = 3.4$ T4C code, or 40–80% above the break-even point, using current state-of-the-art cavity and transmon of the same style as in our experiment. Extrinsic effects such as thermal excitation of the transmon is not included, but in Extended Data Fig. 5b we show for a specific attainable case of $T_{1A} = 1 \text{ ms}$, $T_{1q} = T_{1\phi} = 100 \mu\text{s}$, QEC breakeven can be achieved with a few per cent of transmon thermal populations.

In Extended Data Fig. 5a, assuming constant λ and χ_q , the QEC gain factor decreases at the longest T_{1A} (>2 ms). This is because of the increased

weight of error contribution from off-resonance excitation of the transmon due to imperfect frequency selectivity of the transmon comb. Further reduction of λ (accompanied with smaller reduction of χ_q) is needed to further improve the gain factor. A more promising alternative in this high-coherence regime is to use $|2n+1, f, 0\rangle$ instead of $|2n, e, 0\rangle$ as the first intermediate level of PReSPA, which can immediately reduce all frequency-selectivity-related errors by a factor of 4 at the expense of a 2 $\times$ faster transmon decay (a relatively small contribution in this regime).

Data availability

Data used in this work is available on reasonable request.

Code availability

Code used in this work is available on reasonable request.

47. Grimsom, A. L., Combes, J. & Baragiola, B. Q. Quantum computing with rotation-symmetric bosonic codes. *Phys. Rev. X* **10**, 011058 (2020).
48. Leghtas, Z. et al. Hardware-efficient autonomous quantum memory protection. *Phys. Rev. Lett.* **111**, 120501 (2013).
49. Susskind, L. & Glogower, J. Quantum mechanical phase and time operator. *Phys. Phys. Fiz.* **1**, 49–61 (1964).
50. Michael, M. H. et al. New class of quantum error-correcting codes for a bosonic mode. *Phys. Rev. X* **6**, 031006 (2016).
51. Wang, C. et al. A Schrödinger cat living in two boxes. *Science* **352**, 1087–1091 (2016).
52. Nigg, S. E. et al. Black-box superconducting circuit quantization. *Phys. Rev. Lett.* **108**, 240502 (2012).
53. Werschnik, J. & Gross, E. K. U. Quantum optimal control theory. *J. Phys. At. Mol. Opt. Phys.* **40**, R175–R211 (2007).
54. Glaser, S. J. et al. Training Schrödinger's cat: quantum optimal control. *Eur. Phys. J. D* **69**, 279 (2015).
55. Gollub, C., Kowalewski, M. & de Vivie-Riedle, R. Monotonic convergent optimal control theory with strict limitations on the spectrum of optimized laser fields. *Phys. Rev. Lett.* **101**, 073002 (2008).
56. Khaneja, N., Reiss, T., Kehlet, C., Schulte-Herbrüggen, T. & Glaser, S. J. Optimal control of coupled spin dynamics: design of NMR pulse sequences by gradient ascent algorithms. *J. Magn. Reson.* **172**, 296–305 (2005).
57. Kirchmair, G. et al. Observation of quantum state collapse and revival due to the single-photon Kerr effect. *Nature* **495**, 205–209 (2013).
58. Sun, L. et al. Tracking photon jumps with repeated quantum non-demolition parity measurements. *Nature* **511**, 444–448 (2014).
59. Cahill, K. E. & Glauber, R. J. Density operators and quasiprobability distributions. *Phys. Rev.* **177**, 1882–1902 (1969).

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Author contributions J.M.G. carried out the device design, microwave measurements, and data analysis of the experiment under the supervision of C.W. B.B. generated the numerical pulses for unitary control in the experiment under the supervision of J.K. J.L. fabricated the device and contributed to the cryogenic preparation of the apparatus. S.S. carried out numerical simulations for the experiment. B.B., J.K. and C.W. developed the approximate AQEC theory. C.W. conceived and oversaw this project. J.M.G., B.B., J.K. and C.W. wrote the manuscript with input from all authors.

Competing interests The authors declare no competing interests.

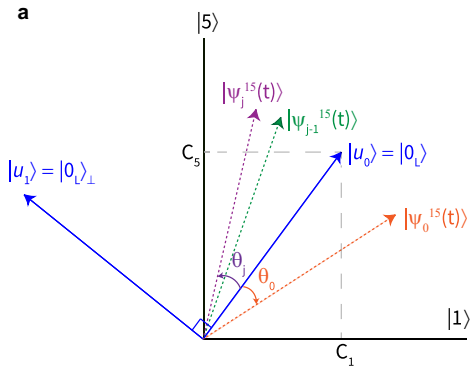
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-021-03257-0>.

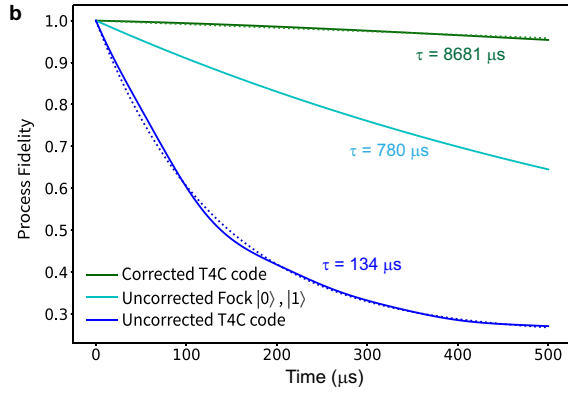
Correspondence and requests for materials should be addressed to C.W.

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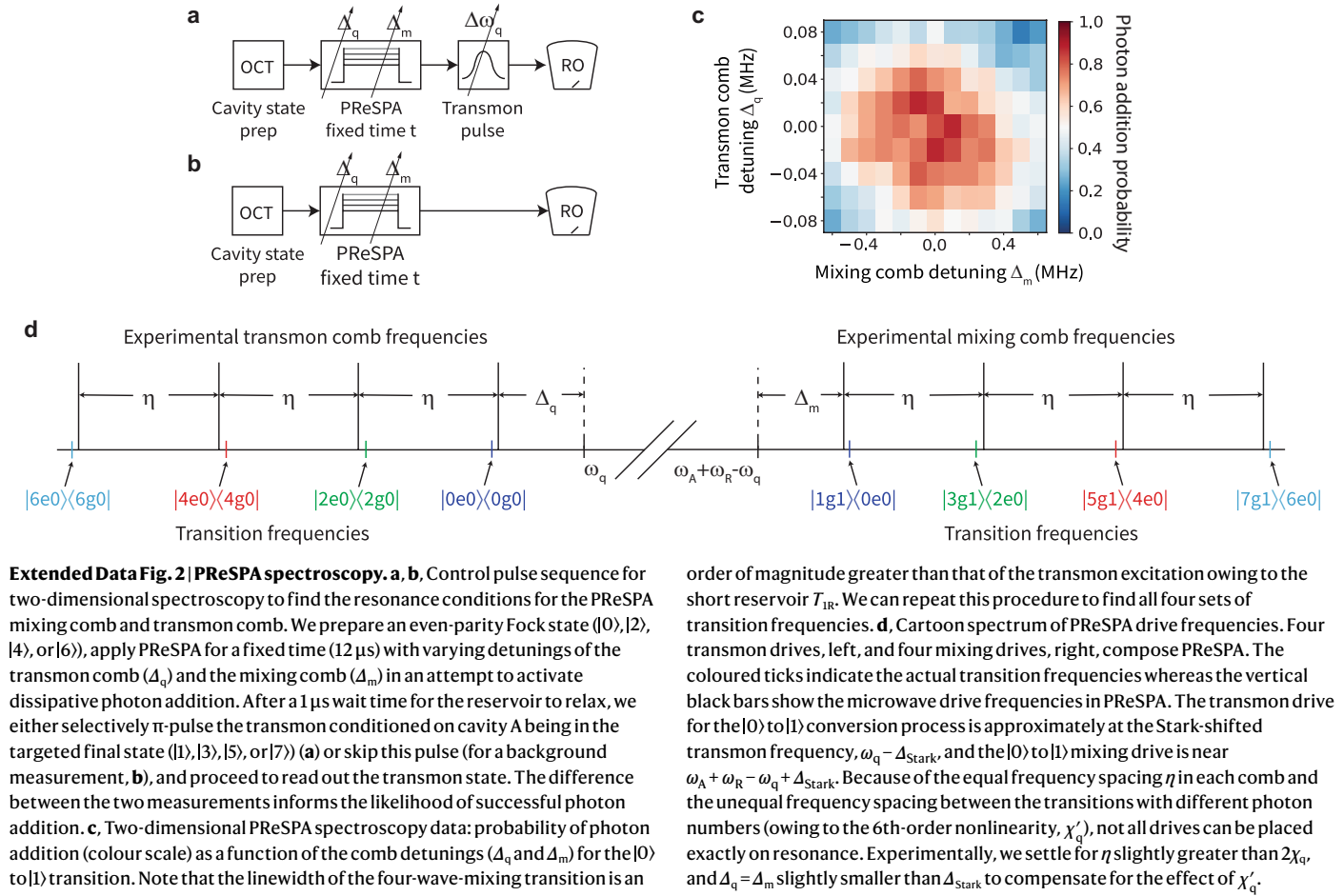
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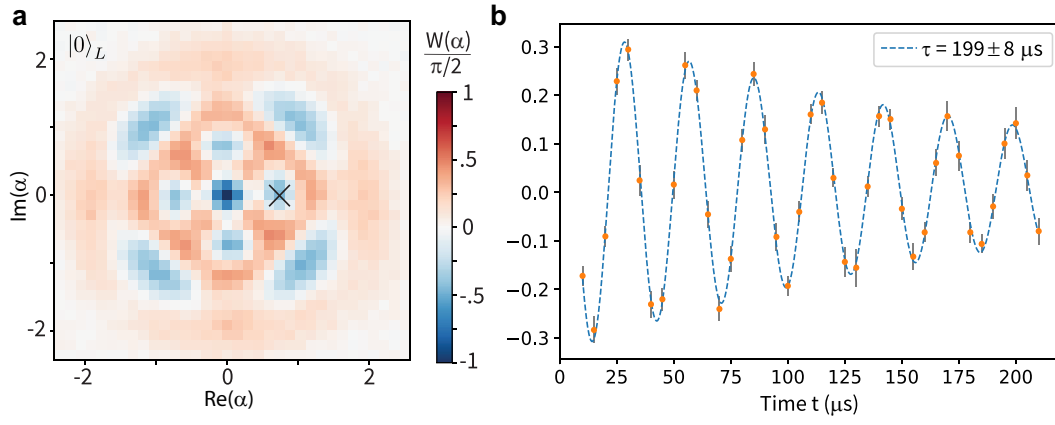


Extended Data Fig. 1 | Dynamics and intrinsic performance of the approximate AQEC protocol. **a**, Diagram illustrating the quantum-trajectory state $|\psi_j^{15}(t)\rangle$ after a time period t under the combined effect of photon loss and instantaneous PReSPA, given a certain number j of jumps. The initial state is $|\psi(0)\rangle = |0_L\rangle$ in the subspace $\mathcal{H}_{15} = \text{span}\{|u_0\rangle, |u_1\rangle\}$. Both jump and no-jump evolution lead to rotation of $|\psi_j^{15}(t)\rangle$ over time within the subspace, and the angle $\theta_j(t)$ parametrizes this rotation. For each quantum trajectory, $|\psi_j^{15}(t)\rangle$ slowly and continuously rotates clockwise in the absence of jumps and occasionally undergoes stochastic jumps counter-clockwise. The diagram and



the dynamics for the states $|\psi_j^{37}(t)\rangle$ in the $\mathcal{H}_{37} = \text{span}\{|u_0\rangle, |u_1\rangle\}$ subspace (not shown) follows an analogous pattern. **b**, Comparing the decay of process fidelities for three cases: T4C encoding using the ideal PReSPA scheme of this section (corrected T4C code, green), T4C encoding without using PReSPA (uncorrected T4C code, teal), and Fock state encoding (uncorrected Fock $|0\rangle, |1\rangle$, blue). Experimental values of T_{14} and K are used, and cavity dephasing is not considered. Exponential curves for the T4C fidelity use the equation $\mathcal{F}_{\text{process}}(t) = 0.75e^{-t/\tau} + 0.25$ to extract decay rate τ .

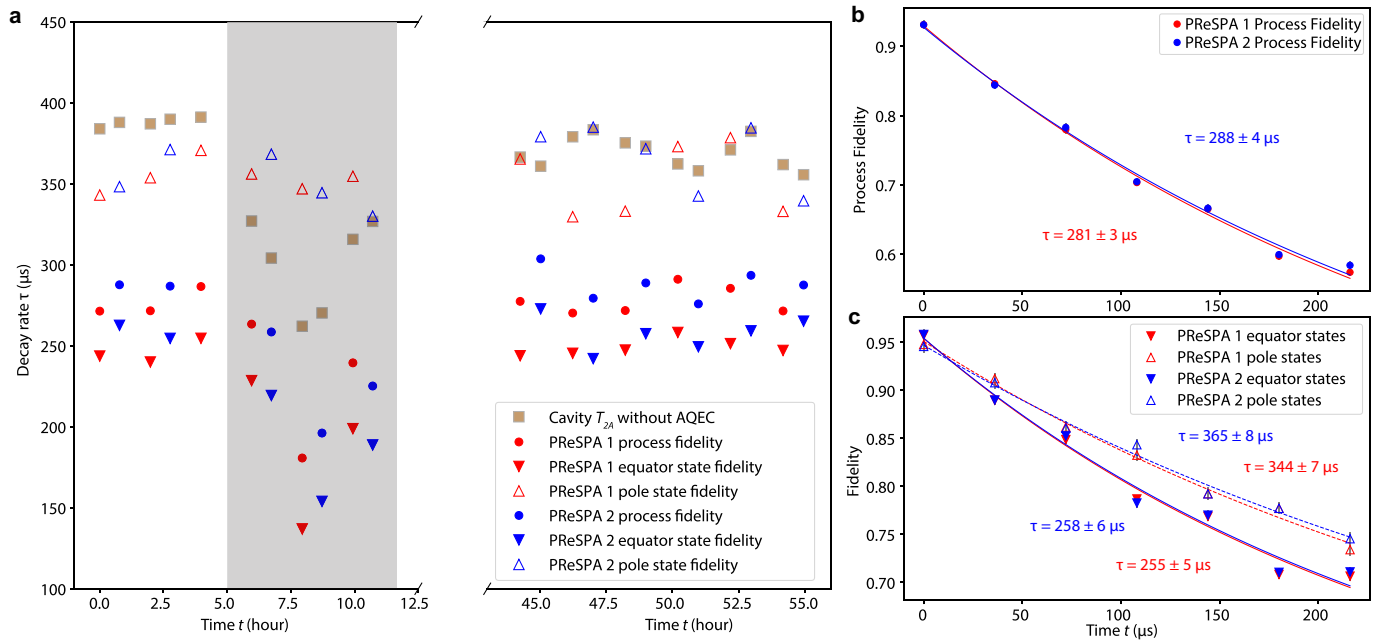




Extended Data Fig. 3 | Cavity Wigner and PReSPA Ramsey measurements.

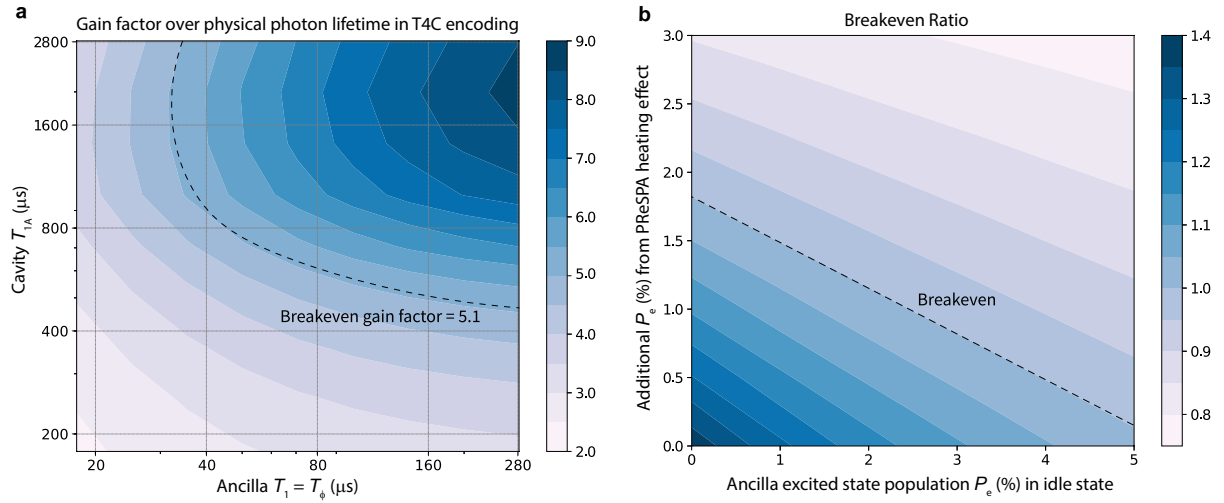
a, Experimental Wigner function $W(\alpha)$ (colour scale, dimensionless) of $|0\rangle_L$, acquired by applying a cavity displacement operation $\hat{D}_\alpha = \exp(a\hat{a}^\dagger - \alpha^* \hat{a})$ with variable complex amplitude α followed by an ancilla-assisted photon-number-parity measurement (which is composed of two $\pi/2$ pulses of the ancilla and a delay time of π/χ_q and an ancilla readout^{37,58}). The Wigner function rotates around the origin over time at a rate proportional to the frequency difference between $|1\rangle$ and $|5\rangle$ in the rotating frame of the experiment. **b**, Measured Wigner function values at a fixed phase-space position (as indicated by the cross in **a**, at $\alpha = 0.75$) as a function of time under PReSPA. Analogous to a qubit Ramsey

measurement, this cavity PReSPA Ramsey experiment can be used to efficiently track the phase evolution of any two-component superposition states using the interference effect enabled by the coherent cavity displacement ($\hat{D}_\alpha$) before readout. The exponential envelope of the sinusoidal fit indicates the rate of decay for the coherence between $|1\rangle$ and $|5\rangle$ under the correction of PReSPA. Similar measurements are applied to various superposition states to provide direct calibration of the frequencies and phases of these states under PReSPA. PReSPA enhances the ability to use such Ramsey measurements at high photon numbers because it approximately preserves photon number distributions in the cavity.



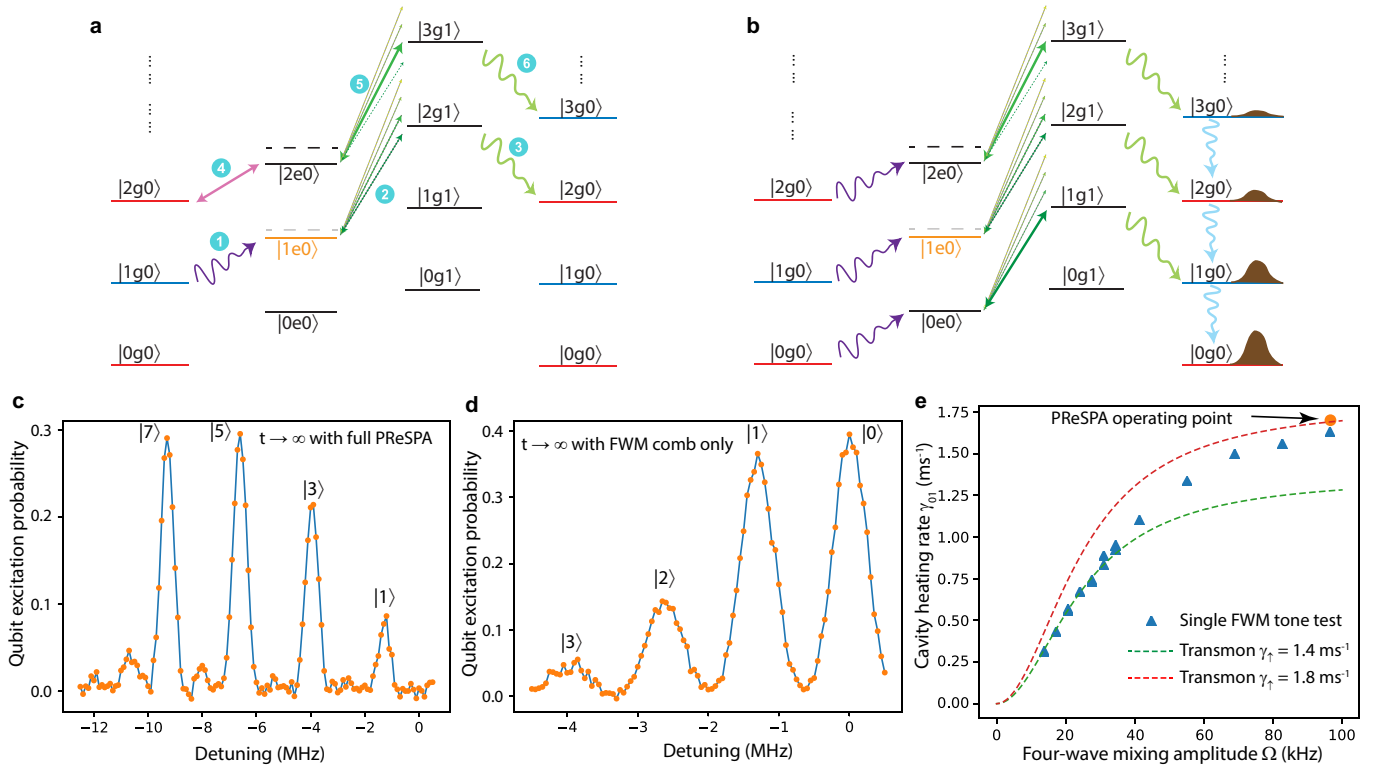
Extended Data Fig. 4 | Tracking AQEC performance over time. **a**, We interleave measurements of the corrected logical qubits under two differently calibrated PReSPA parameter sets (PReSPA-1, red and PReSPA-2, blue) while also monitoring the cavity T_{2A} . Each circle corresponds to the decay time of process fidelity extracted from measuring all six cardinal points of the logical Bloch sphere as described in Fig. 4. The decay rates for state fidelity are shown in triangles for the two logical pole states (upwards triangles) and the four equator states (downwards triangles). The state fidelity is measured by quantum state tomography of the ancilla after decoding the cavity state as described in the Methods. For ancilla state tomography, we measure all six Pauli operators ($\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z, -\hat{\sigma}_x, -\hat{\sigma}_y, -\hat{\sigma}_z$) by performing ancilla rotations before readout. The over-complete measurement set is used for simultaneous calibration of the readout signal contrast, allowing for accurate determination of the transmon state. PReSPA-1 is calibrated by adjusting control parameters

to achieve matched PReSPA rates and zero conversion phases as discussed in the Methods. For PReSPA-2, we employ further empirical parameter optimization to maximize equator state lifetime as described in the Supplementary Information section 5. Cavity A has a two-state switching behaviour of unknown origin (see notes in Extended Data Table 1). For distinct stretches of 2–8 h, cavity A shows fluctuating and abnormally low T_{2A} , and all data recorded during such periods (with example data shown in the shaded region) are excluded in all other parts of the paper. **b**, Process fidelity averaged over the data, excluding the shaded region, for both PReSPA 1 and 2. Data reported in Fig. 4d for the corrected T4C encoding is a duplication of the blue points here. **c**, Equator and pole state fidelity for the same time period for both PReSPA 1 and 2. Oscillatory behaviour in the data are caused by the numerical differences between the two decoding pulses discussed in Methods section ‘GRAPE methods’.



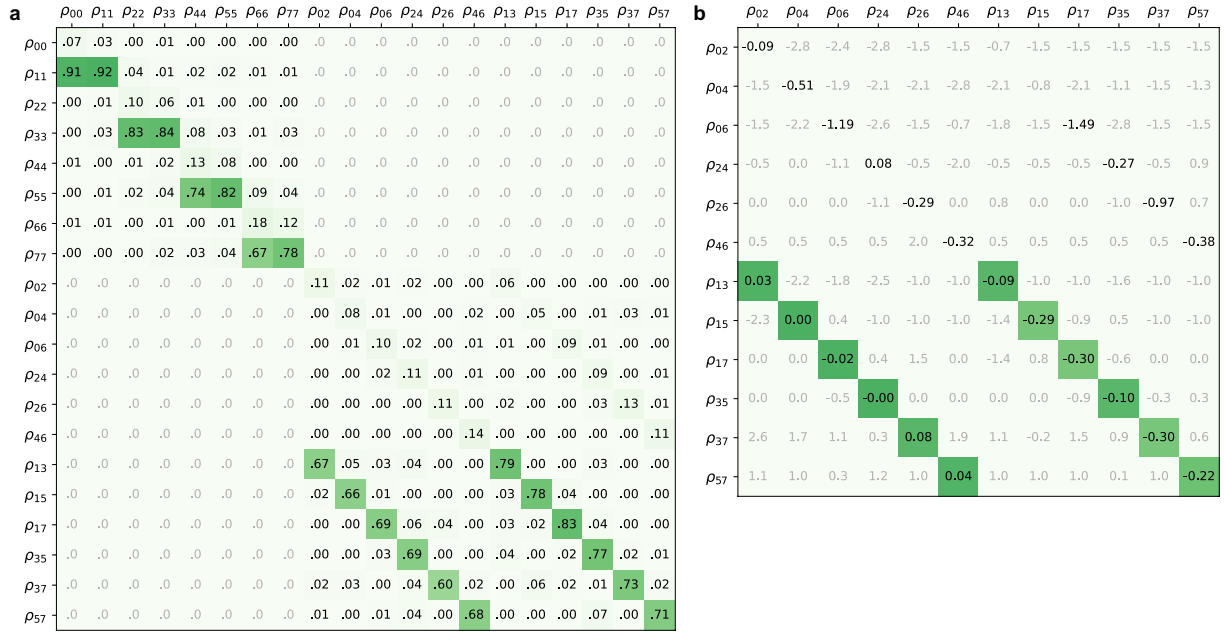
Extended Data Fig. 5 | Predicted AQEC performance in numerical simulations. The results are based on master-equation simulations of a T4C qubit (with encoding $\bar{n} = 3.4$ for both basis states) under the Hamiltonian equations (11) and (18), which captures the dynamics under the microwave combs of PreSPA. **a**, Gain factor (colour scale, dimensionless) of the corrected logical qubit lifetime over the physical photon lifetime ($T_{1A}/\bar{n}$) in the T4C encoding as a function of ancilla T_{1q} , T_ϕ (which are made equal for convenience) and cavity T_{1A} . To illustrate the intrinsic performance of our transmon-based PreSPA pumping scheme, we have assumed no ancilla thermal excitations and other cavity dephasing errors. However, ancilla excitations due to the imperfect frequency selectivity of PreSPA, which is unrelated to photon loss, are reflected in the simulation. Therefore, the gain factor shown is different from the G_i defined in the main text, and decreases slightly at long T_{1A} . **b**, The QEC break-even ratio (colour scale, dimensionless), defined as the T4C qubit lifetime under PreSPA over the lifetime of the 0/1 Fock-state encoding. Here we

use a specific set of achievable coherence times $T_{1A} = 1 \text{ ms}$, $T_{1q} = T_q = 100 \mu\text{s}$ (refs. ^{46,51}) and show the degradation of AQEC performance in the presence of spontaneous transmon excitation (γ_s) errors caused by the stray thermal background (horizontal axis) or pump-tone-induced heating from PreSPA (vertical axis). QEC breakeven can be reached if the γ_s rate is kept reasonably low. The dashed lines in both **a** and **b** indicate where the QEC break-even ratio equals 1. Relevant system parameters: In **a**, we use $\lambda/2\pi = 17.5 \text{ kHz}$, $\Omega/2\pi = 45 \text{ kHz}$, $\kappa/2\pi = 227 \text{ kHz}$, $\chi_q/2\pi = 1.05 \text{ MHz}$, $\chi_R/2\pi = 1.6 \text{ MHz}$, scaled from the experiment by 50%, 50%, 40%, 80%, 80% respectively. In **b**, we use $\lambda/2\pi = 21.6 \text{ kHz}$, $\Omega/2\pi = 72 \text{ kHz}$, $\kappa/2\pi = 364 \text{ kHz}$, $\chi_q/2\pi = 1.18 \text{ MHz}$, $\chi_R/2\pi = 1.8 \text{ MHz}$, scaled from the experiment by 80%, 80%, 64%, 90%, 90% respectively. The choice of parameters here is guided by the scaling laws of various error channels (Extended Data Table 3) but did not go through optimization of individual parameters.



Extended Data Fig. 6 | Cavity heating effect caused by spurious transmon excitations. **a**, Schematic mechanism of sequential two-photon gain triggered by spurious excitation of the transmon. Starting from $|1g0\rangle$, following the transitions labelled with blue-circled numbers, the system is excited to $|1e0\rangle$ by a transmon $\gamma_\uparrow$ jump, and then driven unintentionally to $|2g1\rangle$ by mixing tones that are off-resonance by only $\pm\chi_q/2\pi = \pm 1.3$ MHz (which does not provide strong enough frequency selectivity relative to the reservoir linewidth $\kappa/2\pi = 0.58$ MHz), and then relaxes to $|2g0\rangle$ following reservoir decay. Once a photon is added in this spurious odd-to-even conversion process, the PReSPA scheme by design will add a second photon, driving the system ultimately to $|3g0\rangle$. Similarly, a transmon $\gamma_\uparrow$ jump can add two photons to $|3\rangle$ and $|5\rangle$ states (but not $|7\rangle$). **b**, Schematic illustration of the steady-state photon number distribution established between the ancilla- $\gamma_\uparrow$ -induced photon addition and the natural photon loss of the cavity. The figure corresponds to the configuration of the test experiment in **d** when only the mixing comb (but not the transmon excitation comb) is applied. **c**, Steady-state ($t \rightarrow \infty$) cavity photon number distribution under PReSPA, as probed by ancilla qubit spectroscopy.

In this measurement, PReSPA is applied nearly at all times, only briefly interrupted by spectroscopy probes once every 2 ms. The peak amplitudes confirm that odd photon number parity is permanently stabilized, and also reveals the presence of spurious excitation processes in the cavity. **d**, Steady-state cavity photon number distribution when only the mixing comb of PReSPA is applied, as in **b**. Under this configuration, a transmon $\gamma_\uparrow$ jump may add just one spurious photon in the cavity, leading to an effective cavity heating rate out of its vacuum state $\gamma_{01} \approx \gamma_\uparrow$ (when the transmon decay rate through the reservoir $\gamma_m \gg 1/T_{1q}$). The relative probability of $|0\rangle$ versus $|1\rangle$ informs the balance between the cavity decay rate $1/T_{1a} = 1.9$ ms $^{-1}$ and the heating rate γ_{01} . **e**, Blue triangles show the cavity heating rate γ_{01} , as measured with the technique described in **d**, as a function of the Rabi amplitude of a single $|0e0\rangle \leftrightarrow |1g1\rangle$ mixing tone. γ_{01} initially follows the expected values corresponding to transmon $\gamma_\uparrow = 1.4$ ms $^{-1}$ (green dashed curve), but additional heating effects due to the microwave pump is observed at higher four-wave mixing (FWM) rate, Ω .



Extended Data Fig. 7 | Process χ matrix block for 25 μ s of PReSPA. The matrix converts elements of input density matrices (top axis) to output density matrix elements (left axis) expressed in the Fock state basis. **a**, Amplitude values of the χ matrix elements. The upper left block ($\chi_{nm,mm}$) describes the conversion of diagonal elements of the input and output density matrices, which is associated with transfer of photon occupation probabilities calculated from transmon spectroscopy experiments (as in Fig. 2d, e). The lower right block ($\chi_{nm,kl}$) describes the conversion of relevant off-diagonal elements of density matrices, which is calculated from Wigner tomography and density matrix reconstruction⁵⁹ (as in Fig. 3). The greyed blocks are assumed to be zero owing to the absence of interference between the four conversion paths in PReSPA. **b**, Phase values of the χ matrix elements for the lower right block in **a**. For best

illustration of the PReSPA process, the phases are reported in the full rotating frame where all Fock states have zero energy. Values in grey are measured but not statistically significant because the corresponding amplitude value is not large enough. In this frame, as prescribed by equation (2), PReSPA requires zero phase for the six $\chi_{nm,(n+1)(m+1)}$ elements representing the coherence of the even-to-odd conversion process, which is accomplished by our PReSPA calibration. The diagonal elements $\chi_{nm,mm}$ representing the preservation of odd-state superpositions should have zero phase by definition. Their systematic deviation from zero was caused by parameter drift in the experiment as that block of data was acquired at a later time than the earlier rotating frame calibration.

Extended Data Table 1 | System parameters

	Symbol	Value
Transmon frequency	$\omega_q/2\pi$	5489.59 MHz
Transmon anharmonicity	$\alpha_q/2\pi$	201.22 MHz
Transmon T_1	T_{1q}	39 μ s
Transmon T_2^* Ramsey	T_{2q}^*	17 μ s [†]
Transmon T_2 Echo		36 μ s
Transmon $ e\rangle_q$ population		5%
Reservoir frequency	$\omega_R/2\pi$	7337.8 MHz
Reservoir-transmon coupling	$\chi_R/2\pi$	2.8 MHz
Reservoir T_1	$1/\kappa$	0.27 μ s
Cavity A frequency	$\omega_A/2\pi$	4067.445 MHz
Cavity A-transmon coupling	$\chi_q/2\pi$	1.313 MHz
Cavity A-reservoir coupling	$\chi_{AR}/2\pi$	7 kHz [‡]
Cavity A anharmonicity	$K/2\pi$	1.7 kHz
Cavity A 2nd order coupling	$\chi_q'/2\pi$	5.5 kHz
Cavity A T_1	T_{1A}	520 μ s
Cavity A T_2	T_{2A}	380 μ s [#]
Cavity A $ 1\rangle$ population		$\sim$ 1%

[†]The transmon T_2^* reflects any 1/e decay time of Ramsey oscillations. The transmon displays a random switching behaviour between two values of $\omega_q/2\pi$ that are 40 kHz apart, with a dwell-time split of approximately 85%:15%. The switching timescale is on the order of subseconds to seconds. All our experimental data reflects the averaged result from sampling the two ancilla frequencies.

[‡]The cross-Kerr χ_{AR} is derived from other measured parameters.

[#]Cavity A has a distinctive switching behaviour between a regular state with stable T_{2A} ($380 \pm 25 \mu$ s) and occasional ‘bad periods’ lasting for 2–8 h where T_{2A} fluctuates wildly in the range of 200–340 μ s. The reduced cavity coherence during these periods is not accompanied by any other changes of system parameters, and can be recovered by a Hahn echo pulse (with echo, cavity $T_{2A} \approx 390 \mu$ s at all times). We exclude data during these unstable periods throughout the paper except for Extended Data Fig. 4 where its impact on AQEC performance is illustrated.

Extended Data Table 2 | Comparison of calculated and measured PReSPA transition rates

Index n	Raw mixing amp. (a.u.)	ξ_n (approx.)	Ω_n (kHz) (theory)	$ \Omega_n $ (kHz) (fit)
0 ($ 0\rangle$ to $ 1\rangle$)	-1.22	-0.058	-125	92
1 ($ 2\rangle$ to $ 3\rangle$)	1.00	0.048	127	88
2 ($ 4\rangle$ to $ 5\rangle$)	0.64	0.030	127	87
3 ($ 6\rangle$ to $ 7\rangle$)	0.49	0.023	124	85

Index n	Raw transmon amp. (a.u.)	Λ_n (kHz)	λ_n (kHz) (theory)	$ \lambda_n $ (kHz) (fit)
0 ($ 0\rangle$ to $ 1\rangle$)	$-0.98 e^{-0.43i}$	$-21.4 e^{-0.43i}$	$-27 e^{-0.37i}$	28
1 ($ 2\rangle$ to $ 3\rangle$)	$1.52 e^{0.00i}$	$33.1 e^{0.00i}$	$28 e^{0.04i}$	27
2 ($ 4\rangle$ to $ 5\rangle$)	$1.27 e^{0.02i}$	$26.7 e^{0.02i}$	$28 e^{0.02i}$	27
3 ($ 6\rangle$ to $ 7\rangle$)	$1.14 e^{-0.35i}$	$24.9 e^{-0.35i}$	$27 e^{-0.34i}$	26

Using calibrated amplitudes and phases of the transmon comb and mixing comb in the experiment, the complex-valued PReSPA transition rates (λ_n and Ω_n) can be calculated based on equations (14) and (15). Here the transmon comb amplitude Λ_n is calibrated from ancilla Rabi oscillations, and the amplitudes of mixing tones are approximately converted to the dimensionless displacement parameter ξ_n by measuring the Stark shift Δ_{Stark} induced by that single tone: $\xi \approx \sqrt{\Delta_{\text{Stark}}/2a_q}$. The experimentally measured PReSPA transition rates result from fitting the time-domain dynamics of the photon addition processes (Fig. 2f, also see Supplementary Fig. 1 for further details), which do not contain phases. Note that Ω_0 is intentionally set opposite to others in phase to suppress multi-tone mixing effects (see equations (14) and (15)) by destructive interference.

Extended Data Table 3 | Breakdown of the decoherence sources of the logical qubit under AQEC

	Rate of occurrence	Longitudinal relaxation rate	Transverse relaxation rate	Rate scaling
Unrelated to single-photon loss:				
ancilla spurious excitation $\gamma_{\uparrow}$	1.8 ms^{-1}	1.9 ms^{-1}	1.8 ms^{-1}	
off-resonant excitation by transmon comb	0.2 ms^{-1}	0.3 ms^{-1}	0.2 ms^{-1}	$< \gamma_m(\lambda/\chi_q)^2$
other cavity dephasing	0.3 ms^{-1}		0.3 ms^{-1}	
Fail to correct single-photon loss:				
ancilla relaxation T_{1q}	$7\% \cdot (\bar{n}/T_{1A})$	0.7 ms^{-1}	0.5 ms^{-1}	$1/(\gamma_m T_{1q}) \cdot (\bar{n}/T_{1A})$
2nd photon decay	$6\% \cdot (\bar{n}/T_{1A})$		0.4 ms^{-1}	$(\bar{n} t_{\text{even-odd}}/T_{1A}) \cdot (\bar{n}/T_{1A})$
incorrect pumping path by mixing comb	$3\% \cdot (\bar{n}/T_{1A})$		0.2 ms^{-1}	$(\kappa/2\chi_q)^2 \cdot (\bar{n}/T_{1A})$
K & correction time uncertainty	$2\% \cdot (\bar{n}/T_{1A})$		0.1 ms^{-1}	$(K t_{\text{even-odd}})^2 \cdot (\bar{n}/T_{1A})$
χ'_q & correction time uncertainty	$1\% \cdot (\bar{n}/T_{1A})$		0.1 ms^{-1}	$(\chi'_q/\gamma_m)^2 \cdot (\bar{n}/T_{1A})$
χ_{AR} & correction time uncertainty	$< 0.5\% \cdot (\bar{n}/T_{1A})$			$(\chi_{AR}/\kappa)^2 \cdot (\bar{n}/T_{1A})$
intrinsic code word distortion	$2.5\% \cdot (\bar{n}/T_{1A})$		0.2 ms^{-1}	
other PReSPA phase errors	$\sim 2\% \cdot (\bar{n}/T_{1A})$		$\sim 0.1 \text{ ms}^{-1}$	
Total		2.8 ms^{-1}	3.8 ms^{-1}	

See Supplementary Information section 3 for detailed discussions on all error channels. Some of the listed decoherence channels can be unambiguously associated with discrete quantum jumps, such as transmon $\gamma_{\uparrow}$, T_{1q} error, 2nd photon loss and other cavity dephasing. Each event fully scrambles cavity photon phases and therefore translates to logical transverse relaxation at a 1:1 ratio. The bit-flip rate from the occurrence of double photon gain (about 50% of $\gamma_{\uparrow}$ events, see Extended Data Fig. 6) and second-photon loss is converted to longitudinal relaxation rate via a factor of 2 by definition. Other decoherence channels are more continuous in nature, for which we define their effective rate of occurrence as the resultant logical transverse relaxation rate. They arise from various competing timescales in PReSPA such as dispersive shift against reservoir linewidth and dissipation rates against various higher-order nonlinearity. The scaling law for each contribution is also listed when applicable. γ_m is the transmon decay rate via the reservoir in the presence of the mixing comb, approximately 0.35 ms^{-1} in our experiment. Definition of other rates can be found in Extended Data Table 1 or in the main text. For intrinsic codeword distortion, see Methods section ‘Approximate AQEC and decoding unitary’. Other cavity dephasing effects are caused by thermal excitation in the reservoir mode and other cavity modes of the system. Other possible small PReSPA phase errors include imperfect matching of λ and Ω for the four photon addition paths, and 2nd-order sensitivity to a low-frequency-noise problem present in our transmon ancilla (see the Extended Data Table 1 footnote).